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Comunicação em IoT – Parte 5

RESIDÊNCIA – Aula Síncrona (05/08/2025)

Prof. Dr Ricardo Menezes Prates

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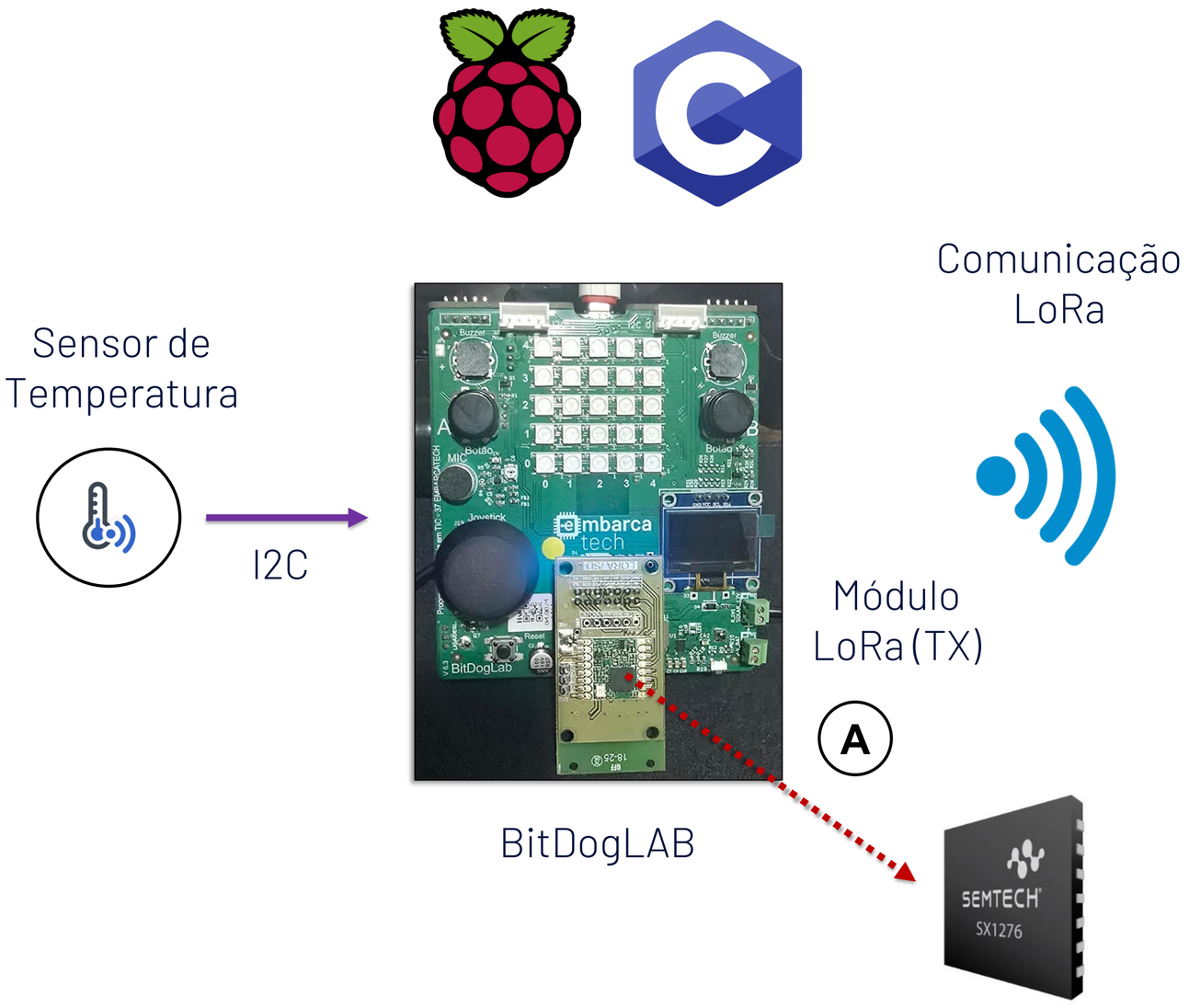
- *Introdução*
  - Planejamento da aula
- Tecnologia de comunicação LoRa
  - *Estrutura do pacote – LoRa Packet Structure*
  - Parâmetros temporais
  - *Transceptor SX1276 e módulo RFM95W*
- Prática – BitDogLAB + Módulo LoRa (RFM95W)
  - *Atividade - Transmissor LoRa – Pico SDK.*
- Conclusões

Introdução LoRa

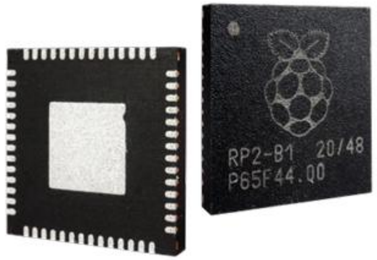
Objetivos

- Avaliar os principais registradores dedicados (LoRa Mode Register Map) presentes no transceptor LoRa **SX1276** e suas formas de uso para configuração de um modulo **RFM95W** através do protocolo SPI.

- Implementar e configurar um transmissor (TX) LoRa através da linguagem C e do kit de desenvolvimento de *software* Pico SDK.



Planejamento da Aula



- Pré-requisitos para esta aula:

✓ Linguagem C e Pico SDK- RP2040;

✓ Portas de Entrada e Saída (GPIO);

✓ *Interface de comunicação* SPI;

✓ *Interfaces de comunicação sem fio;*

✓ *Conhecimento básico da interface LoRa (Parte 4).*

Planejamento da Aula

- Potenciais perguntas a serem trabalhadas nesta aula.

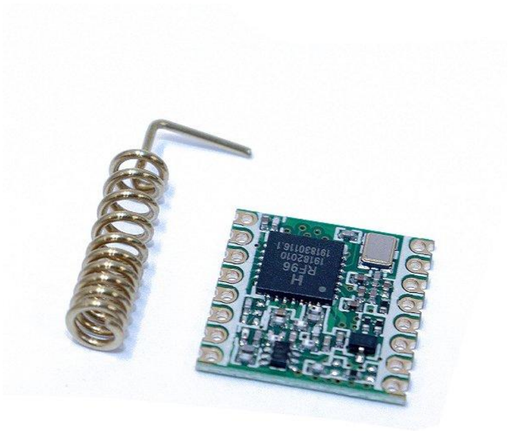
✓ Como é a estrutura de um pacote de dados na comunicação LoRa?

✓ Como posso interagir com os registradores do transceptor SX1276 presente no módulo RFM95W?

✓ Como posso configurar um transmissor (TX) LoRa através da interface SPI do RP2040?

LoRa Mode Register Map

Name (Address)	Bits	Variable Name	Mode	Reset	LoRa™ Description
RegModemConfig1 (0x1D)	7-4	Bw	rw	0x07	Signal bandwidth: 0000 → 7.8 kHz 0001 → 10.4 kHz 0010 → 15.6 kHz 0011 → 20.8kHz 0100 → 31.25 kHz 0101 → 41.7 kHz 0110 → 62.5 kHz 0111 → 125 kHz 1000 → 250 kHz 1001 → 500 kHz other values → reserved In the lower band (169MHz), signal bandwidths 8&9 are not supported)
	3-1	CodingRate	rw	'001'	Error coding rate 001 → 4/5 010 → 4/6 011 → 4/7 100 → 4/8 All other values → reserved In implicit header mode should be set on receiver to determine expected coding rate. See 4.1.1.3
	0	ImplicitHeaderModeOn	rw	0x0	0 → Explicit Header mode 1 → Implicit Header mode



LoRa

LoRa

- Parâmetros da tecnologia LoRa - revisão:

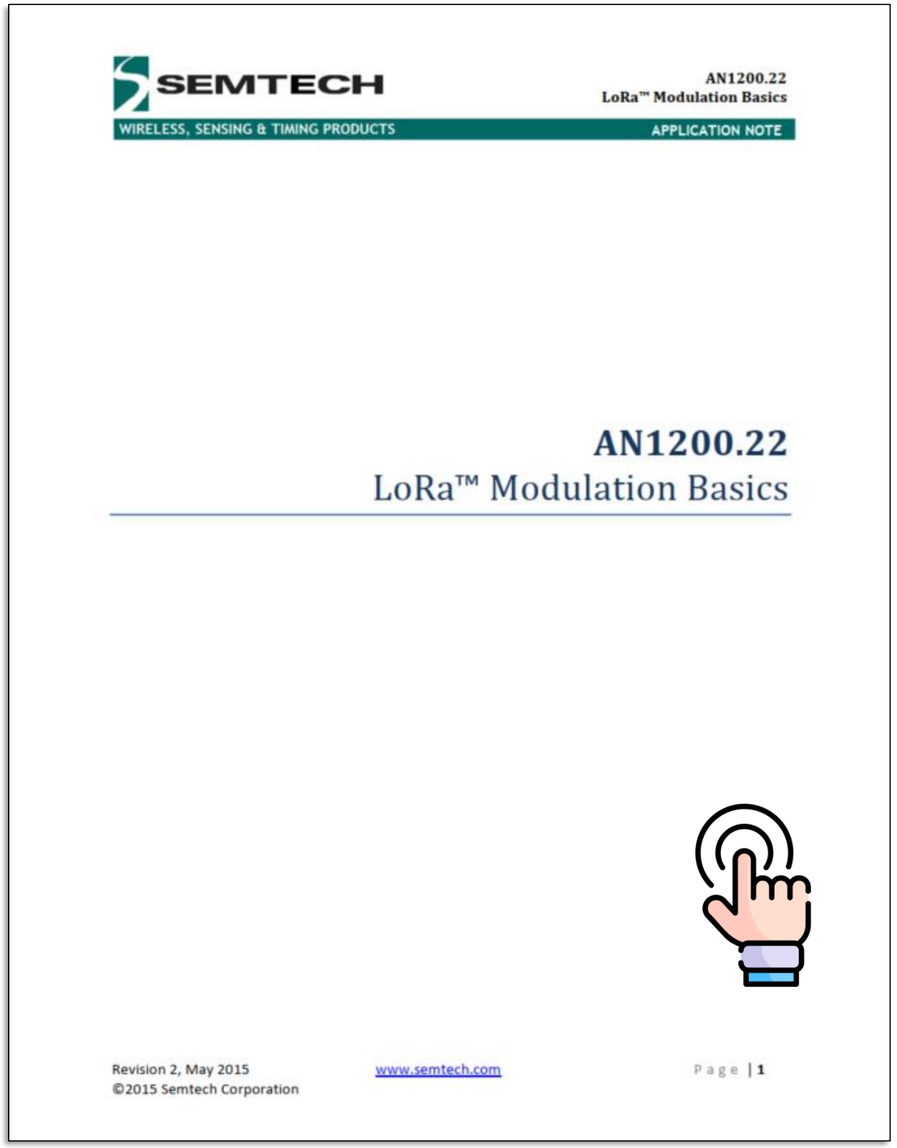
✓ Fator de Espalhamento (*Spreading Factor* - SF);

✓ Potência de Transmissão (*Transmit Power* - TP);

✓ Taxa de Codificação (*Coding Rate* – CR);

✓ Taxa de Modulação de bits (*Data Rate* – Rb);

✓ Largura de Banda (*Modulation Bandwidth* – BW).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APPLICATION NOTE

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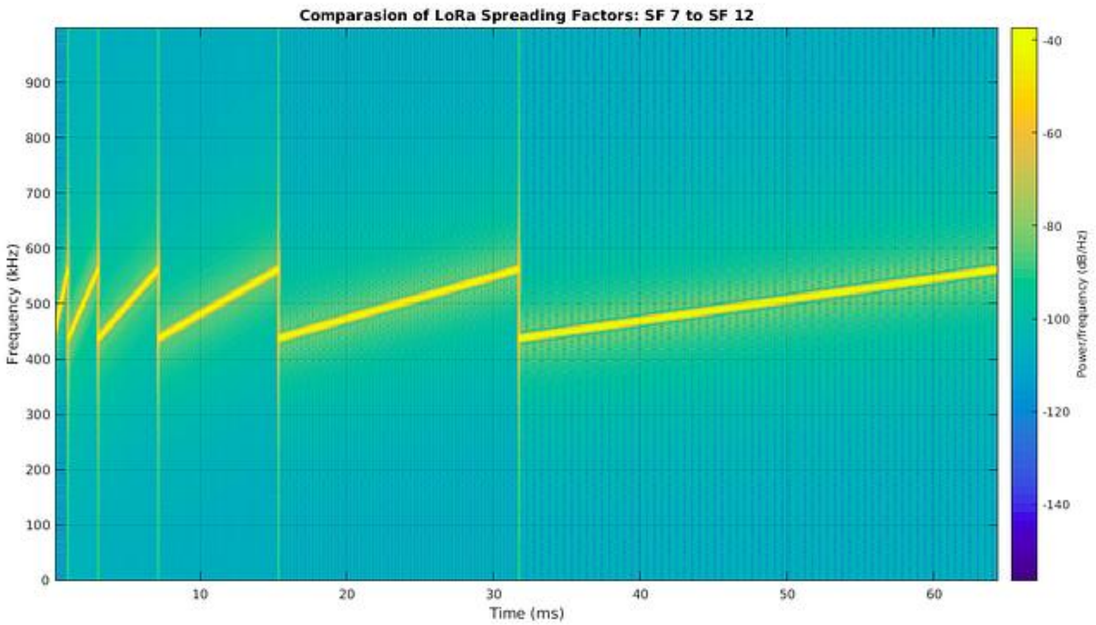
Parâmetros LoRa

- Taxa de modulação de bits – Taxa de transmissão do símbolo (Rs) - revisão:

$$T_s = \frac{2^{SF}}{BW}\ seg \qquad R_s = \frac{BW}{2^{SF}}\ simb/seg$$



T_S-> *symbol period (seg);*

R_b-> *bit rate (bps);*

SF-> *spreading fator (7..12);*

CR-> *code rate (1..4);*

BW-> *modulation bandwidth (Hz).*

LoRa

- Taxa de modulação de bits – revisão:

.

$$R_b = SF \times \frac{\left[\frac{4}{4+CR}\right]}{\left[\frac{2^{SF}}{BW}\right]}$$

R_b-> *bit rate (bps);*

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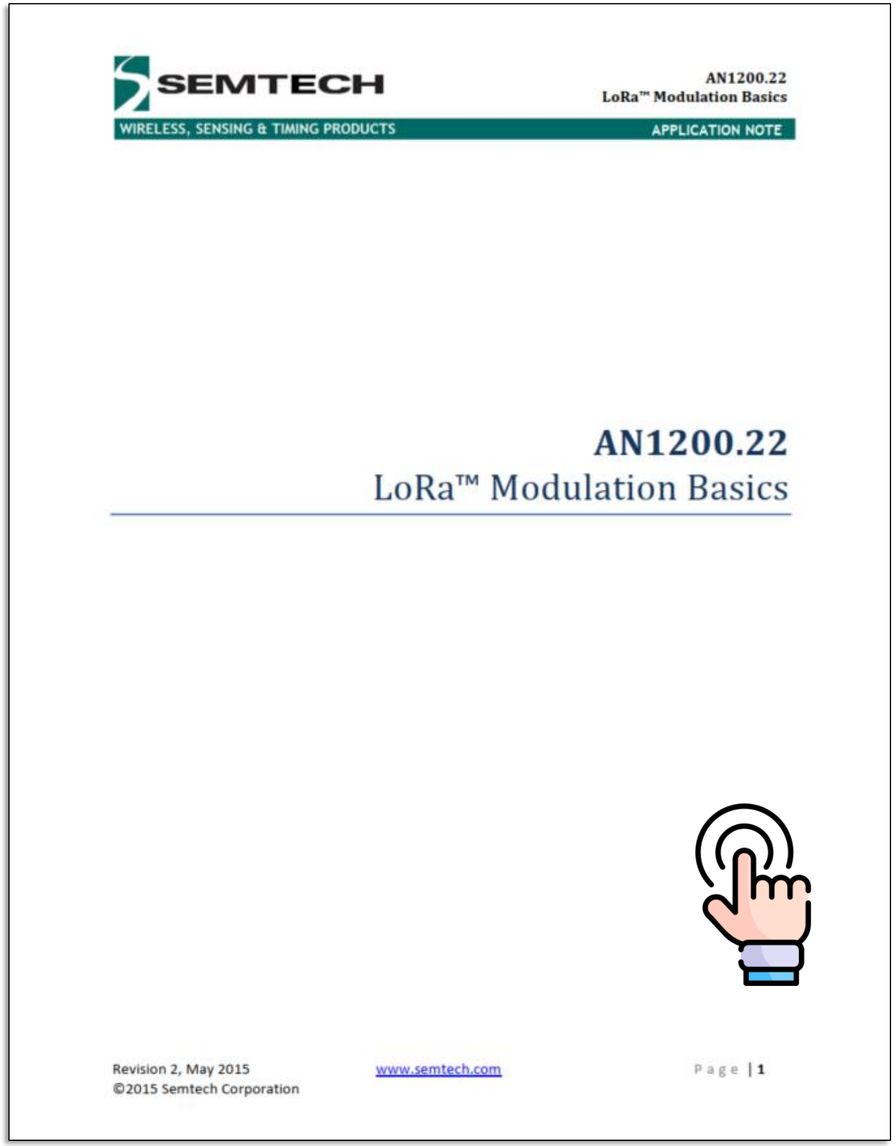
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LoRa

- Exemplo de um sinal LoRa - Taxa de Modulação de bits (*Data Rate*) - revisão:

.

$$R_b = SF \times \frac{\left[\frac{4}{4+CR}\right]}{\left[\frac{2^{SF}}{BW}\right]} = 7 \times \frac{\left[\frac{4}{4+1}\right]}{\left[\frac{2^7}{125.000}\right]} = 5.468{,}75 \approx 5{,}47kbps$$

R_b-> *bit rate (bps);*
SF-> *spreading fator (7);*
CR-> *code rate (1);*
BW-> *modulation bandwidth (125 KHz).*

Dica – Calculadora de taxa de bits e sensibilidade LoRa: https://unsigned.io/understanding-lora-parameters/

LoRa

- Parâmetros adicionais da tecnologia LoRa – SX1276:

✓ Preâmbulo (*Preamble*);

✓ *Header*;

✓ *Payload*;

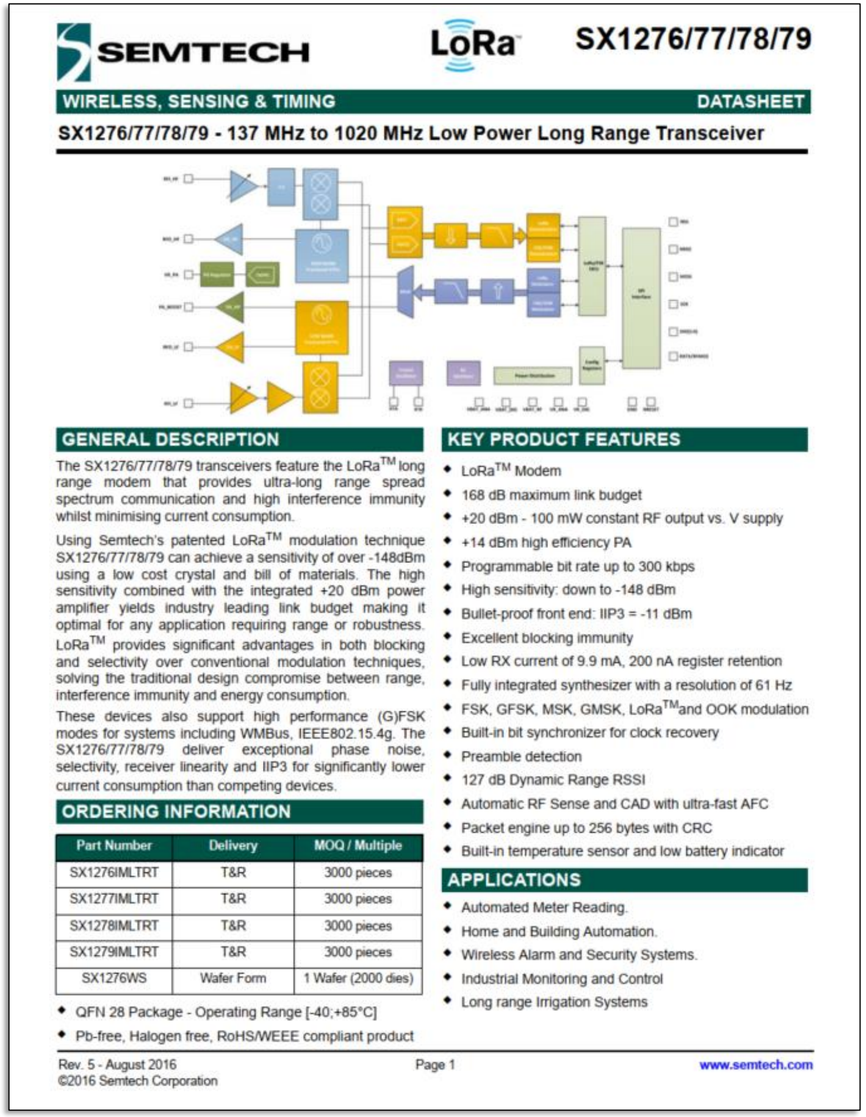
✓ Modo Explícito/Implícito (Explicit Header Mode);

✓ CRC;

✓ *Low Data Rate Optimization*;

✓ *Frequency Hopping*.

✓ Tempo no Ar (*Time on Air* – ToA);



SEMTECH LoRa SX1276/77/78/79

WIRELESS, SENSING & TIMING DATASHEET

SX1276/77/78/79 - 137 MHz to 1020 MHz Low Power Long Range Transceiver

GENERAL DESCRIPTION

The SX1276/77/78/79 transceivers feature the LoRa™ long range modem that provides ultra-long range spread spectrum communication and high interference immunity whilst minimising current consumption.

Using Semtech's patented LoRa™ modulation technique SX1276/77/78/79 can achieve a sensitivity of over -148dBm using a low cost crystal and bill of materials. The high sensitivity combined with the integrated +20 dBm power amplifier yields industry leading link budget making it optimal for any application requiring range or robustness. LoRa™ provides significant advantages in both blocking and selectivity over conventional modulation techniques, solving the traditional design compromise between range, interference immunity and energy consumption.

These devices also support high performance (G)FSK modes for systems including WMBus, IEEE802.15.4g. The SX1276/77/78/79 deliver exceptional phase noise, selectivity, receiver linearity and IIP3 for significantly lower current consumption than competing devices.

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- QFN 28 Package - Operating Range [-40;+85°C]
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- Automated Meter Reading.
- Home and Building Automation.
- Wireless Alarm and Security Systems.
- Industrial Monitoring and Control
- Long range Irrigation Systems

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Estrutura de Pacote LoRa

- O Pacote LoRa é constituído de três elementos – 4.1.1.6:

✓ Preâmbulo;

✓ Leitor (*header*) opcional;

✓ O *payload* de dados.

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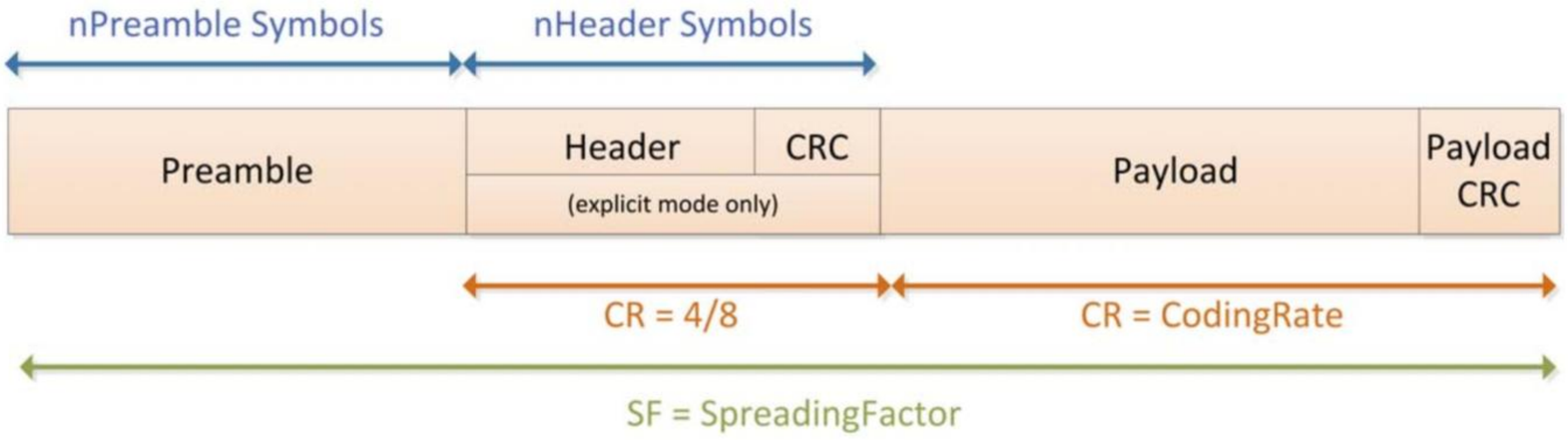


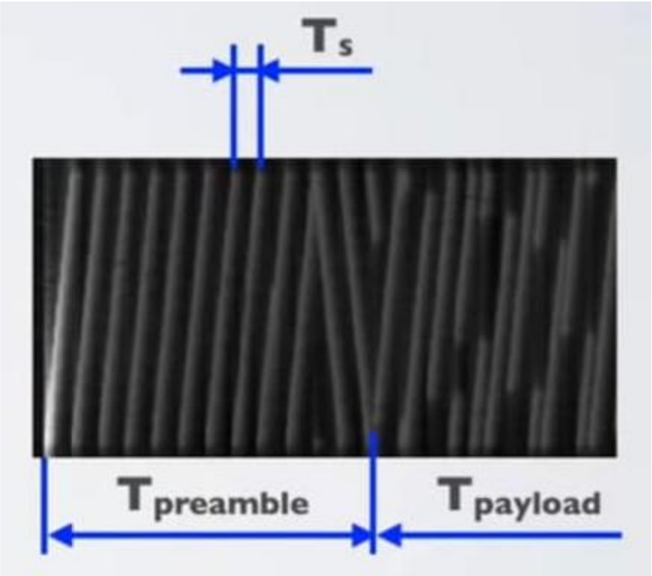
Estrutura de Pacote LoRa

- Preâmbulo:

✓ O preâmbulo é usado para sincronizar o recetor com o fluxo de dados recebidos.

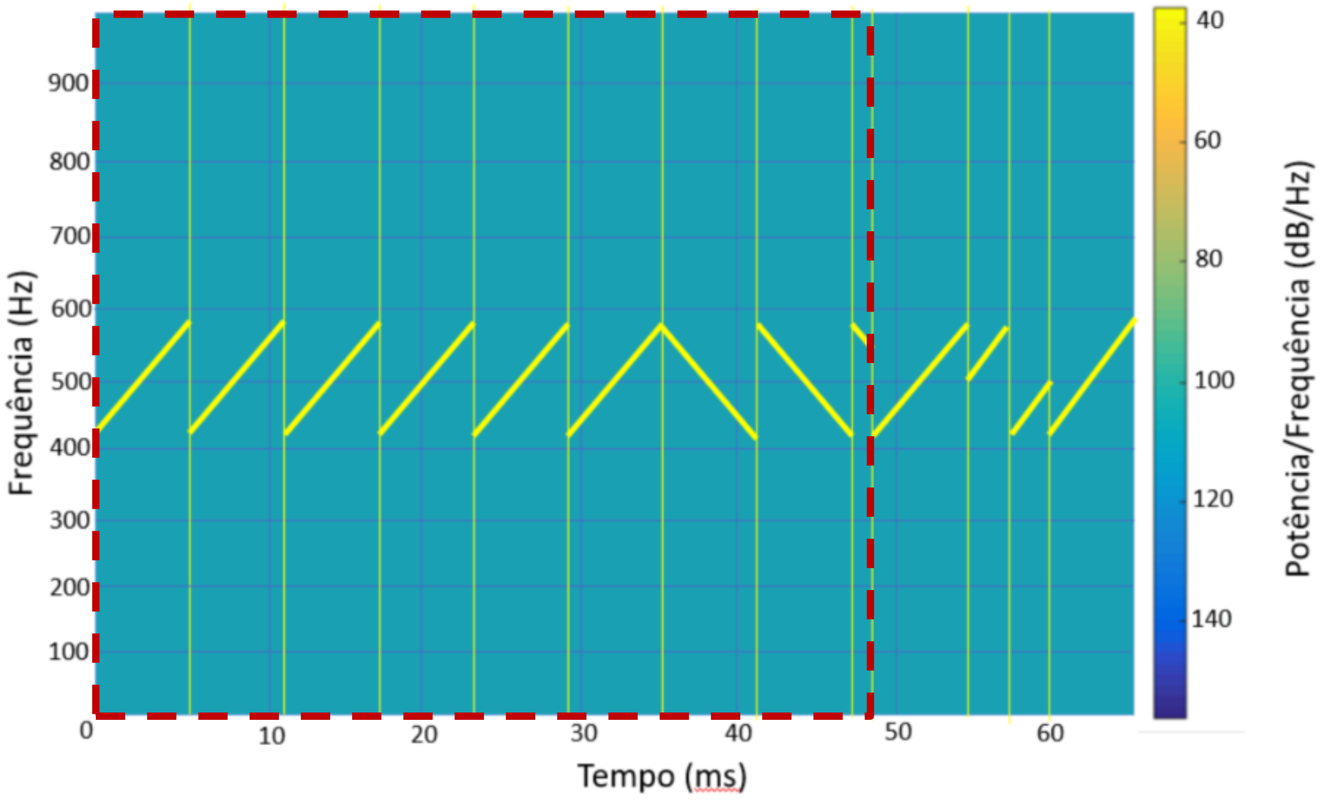
✓ Esta é uma variável programável, o preâmbulo pode ser alterado com o objetivo de reduzir o ciclo de trabalho do receptor em aplicações de recepção intensiva.

✓ O comprimento do preâmbulo transmitido pode ser alterado definindo o registrador *PreambleLength* de 6 para 65535.



Estrutura de Pacote LoRa

- Preâmbulo:



Estrutura de Pacote LoRa

- Preâmbulo:

✓ O receptor (Rx) realiza um processo de deteção de preâmbulo que é reiniciado periodicamente. Por esse motivo, o comprimento do preâmbulo deve ser configurado de forma idêntica ao comprimento do preâmbulo do transmissor.

✓ Quando o comprimento do preâmbulo não é conhecido ou pode variar, o comprimento máximo do preâmbulo deve ser programado no lado do receptor.

Name (Address)	Bits	Variable Name	Mode	Reset	LoRa™ Description
RegPreambleMsb (0x20)	7-0	PreambleLength(15:8)	rw	0x0	Preamble length MSB, = PreambleLength + 4.25 Symbols See Section XX for more details.
RegPreambleLsb (0x21)	7-0	PreambleLength(7:0)	rw	0x8	Preamble Length LSB

Estrutura de Pacote LoRa

- *Header*:

✓ Dependendo do modo de operação escolhido, estão disponíveis dois tipos de cabeçalho (header). O tipo de cabeçalho é selecionado pelo bit ImplicitHeaderModeOn encontrado no registo RegModemConfig1.

✓ O modo explícito apresenta:

- O tamanho do *payload* em bytes;
- O *code rate* (CR) adotado;
- Presença opcional de um CRC de bits para o *payload*.

Name (Address)	Bits	Variable Name	Mode	Reset	LoRa™ Description
RegModemConfig 1 (0x1D)	7-4	Bw	rw	0x07	Signal bandwidth: 0000 → 7.8 kHz 0001 → 10.4 kHz 0010 → 15.6 kHz 0011 → 20.8kHz 0100 → 31.25 kHz 0101 → 41.7 kHz 0110 → 62.5 kHz 0111 → 125 kHz 1000 → 250 kHz 1001 → 500 kHz other values → reserved In the lower band (169MHz), signal bandwidths 8&9 are not supported)
	3-1	CodingRate	rw	'001'	Error coding rate 001 → 4/5 010 → 4/6 011 → 4/7 100 → 4/8 All other values → reserved In implicit header mode should be set on receiver to determine expected coding rate. See Section 4.1.1.3
	0	ImplicitHeaderModeOn	rw	0x0	0 → Explicit Header mode 1 → Implicit Header mode

Estrutura de Pacote LoRa

- *Header – modo explícito*:

✓ O cabeçalho é transmitido com código de correção de erros máximo (4/8). Ele também possui seu próprio CRC para permitir que o receptor descarte cabeçalhos inválidos.

✓ No modo de header explícito, a presença do CRC no final do payload é selecionada apenas no lado do transmissor através do bit RxPayloadCrcOn no registo RegModemConfig1.

Explicit Header	Transmitter	Receiver	CRC Status
Value of the bit RxPayloadCrcOn	0	0	CRC is not checked
	0	1	CRC is not checked
	1	0	CRC is checked
	1	1	CRC is checked

Estrutura de Pacote LoRa

- *Header – modo implícito*:

✓ Em determinados cenários, nos quais o *payload*, a taxa de codificação (CR) e a presença de CRC são fixas ou conhecidas antecipadamente, pode ser vantajoso reduzir o tempo de transmissão invocando o modo de cabeçalho implícito.

✓ Neste modo, o cabeçalho é removido do pacote. O comprimento do payload, o CR e a presença do CRC do payload devem ser configurados manualmente em ambos os lados da ligação de rádio.

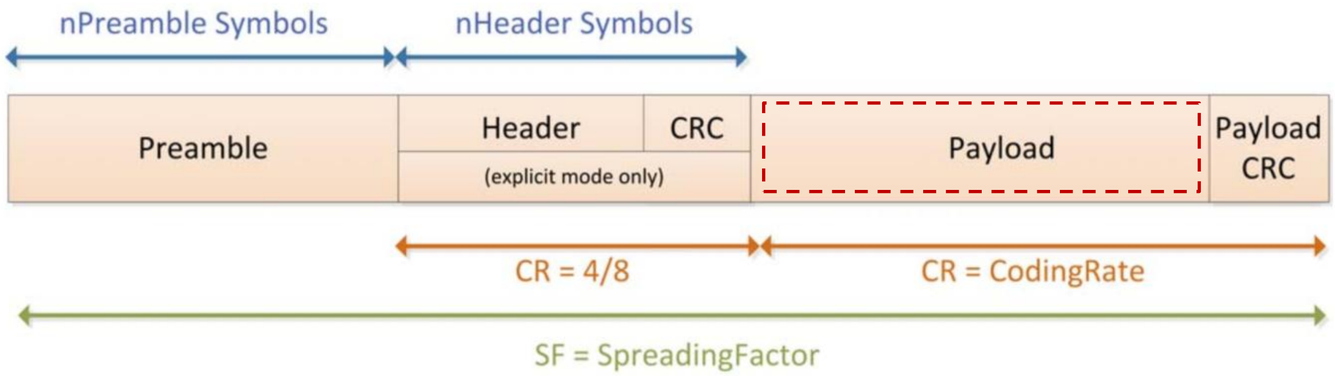
Implicit Header	Transmitter	Receiver	CRC Status
Value of the bit RxPayloadCrcOn	0	0	CRC is not checked
	0	1	CRC is always wrong
	1	0	CRC is not checked
	1	1	CRC is checked

Estrutura de Pacote LoRa

- *Payload*:

✓ É um campo de comprimento variável que contém os dados reais codificados, conforme especificado no cabeçalho no modo explícito ou nas configurações do registo no modo implícito.

✓ Um CRC opcional pode ser anexado.



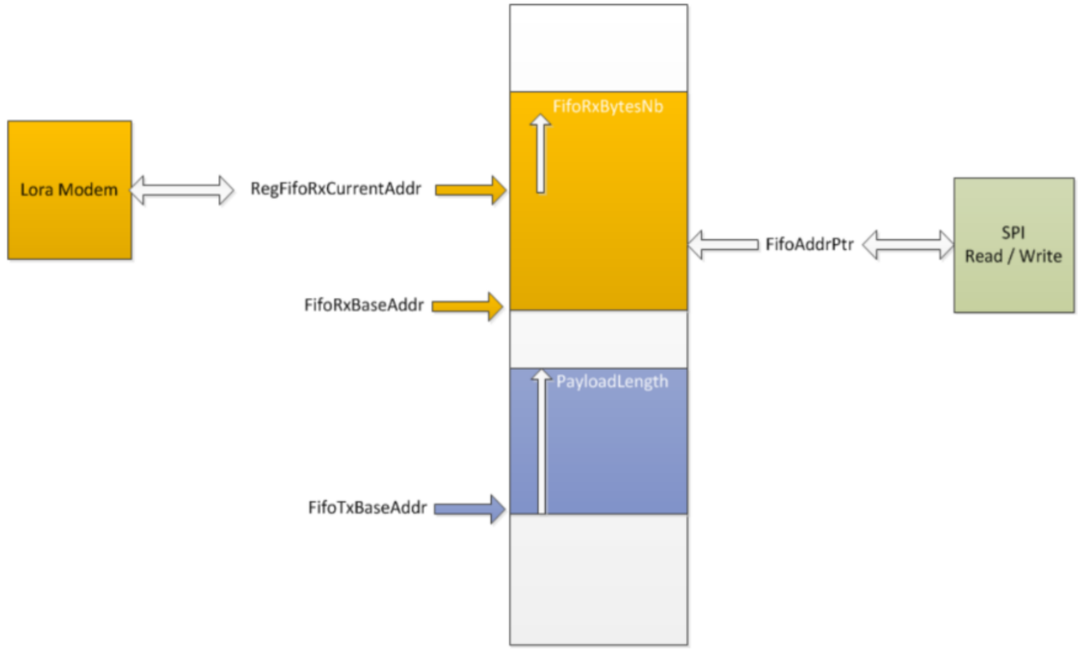
Estrutura de Pacote LoRa

- *Payload – recomendo a leitura do item 4.1.2.3:*

✓ O SX1276 está equipado com um buffer de dados RAM de 256 bytes que é acessível exclusivamente no modo LoRa.

✓ Esta área RAM, aqui referida como buffer de dados FIFO, é totalmente personalizável pelo utilizador e permite o acesso aos dados recebidos ou a serem transmitidos.

✓ Todo o acesso ao buffer de dados FIFO LoRaTM é feito através da interface SPI.



Estrutura de Pacote LoRa

- *CRC (Cyclic Redundancy Check)*:

✓ O CRC é um mecanismo de detecção de erros que é utilizado em comunicações LoRa para garantir a integridade dos dados transmitidos.

✓ Como funciona:

- **Cálculo do CRC**: O transmissor calcula o CRC do pacote de dados antes de transmiti-lo.
- **Inclusão do CRC**: O CRC é incluído no pacote de dados transmitido.
- **Verificação do CRC**: O receptor calcula o CRC do pacote de dados recebido e compara-o com o CRC incluído no pacote.
- **Detecção de erros:** Se o CRC calculado pelo receptor não corresponder ao CRC incluído no pacote, é detectado um erro.

Estrutura de Pacote LoRa

- *Low Data Rate Optimization:*

✓ Dada a duração potencialmente longa do pacote com SF elevados, é oferecida a opção de melhorar a robustez da transmissão em relação às variações de frequência durante a duração da transmissão e recepção do pacote.

✓ O bit LowDataRateOptimize aumenta a robustez da ligação LoRa nessas baixas taxas de dados efetivas.

✓ A sua utilização é obrigatória quando a duração do símbolo excede 16 ms. Observe que tanto o transmissor quanto o receptor devem ter a mesma configuração para LowDataRateOptimize.

Estrutura de Pacote LoRa

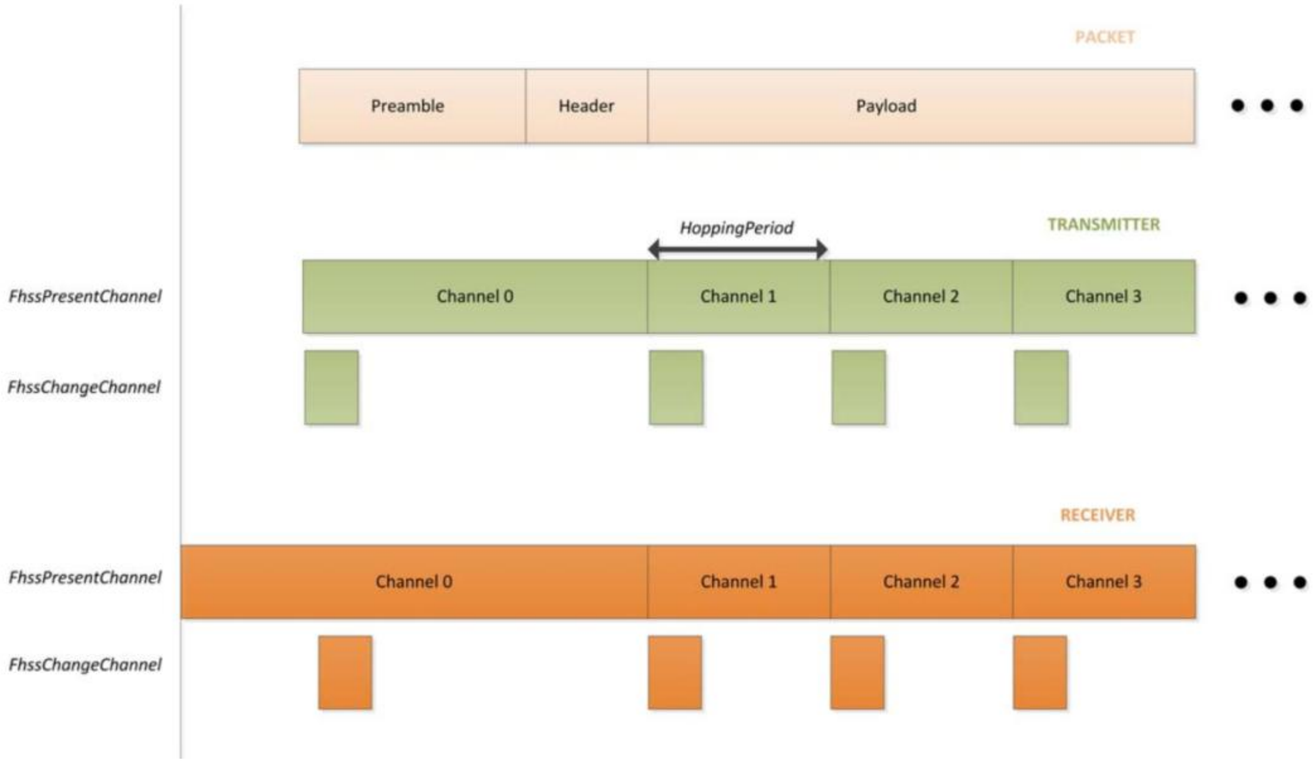
- *Frequency Hopping - 4.1.1.8.:*

✓ O espectro espalhado por salto de frequência (FHSS) é normalmente utilizado quando a duração de um único pacote excede os requisitos regulamentares relativos ao tempo máximo permitido de permanência no canal.

✓ Para facilitar a implementação do FHSS, o modo de salto de frequência do modem LoRaTM pode ser ativado definindo FreqHoppingPeriod para um valor diferente de zero no registo RegHopPeriod.

Estrutura de Pacote LoRa

- *Frequency Hopping:*



Estrutura de Pacote LoRa

- *Frequency Hopping - benefícios:*

- Melhoria da resistência a interferências: A mudança de frequência ajuda a evitar interferências e ruídos.

- Aumento da segurança: A aleatoriedade da mudança de frequência torna mais difícil para interceptar os dados.

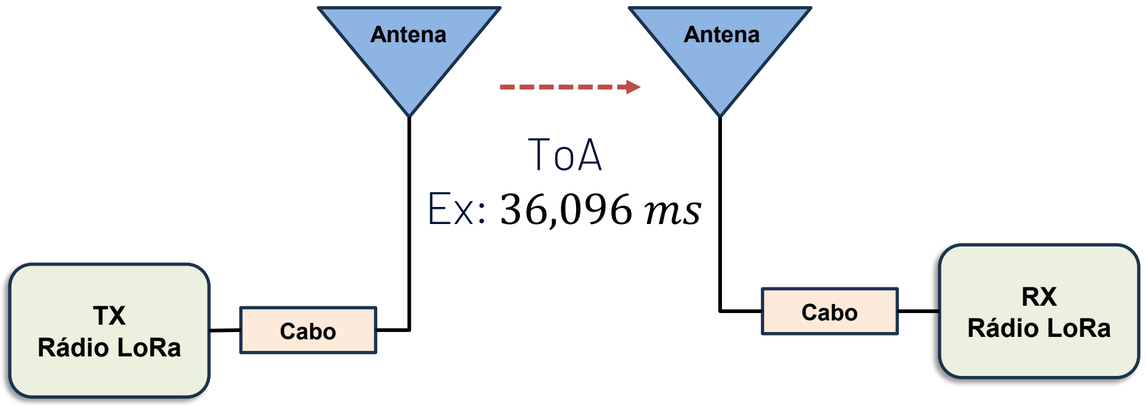
- Melhoria da confiabilidade: O *Frequency Hopping* pode ajudar a garantir que os dados sejam transmitidos de forma confiável e segura.

Parâmetros temporais

- *Time on Air (ToA)*

✓ O ToA (Time on Air) é um parâmetro de comunicações LoRa que se refere ao tempo que um pacote de dados leva para ser transmitido pelo ar.

✓ O ToA afeta a eficiência do uso do espectro e a capacidade de transmissão de dados. Um ToA mais longo pode reduzir a capacidade de transmissão de dados e aumentar a probabilidade de colisões de pacotes.



Parâmetros temporais

- Time on Air (ToA) – parâmetros (pag. 31 – 4.1.1.7):

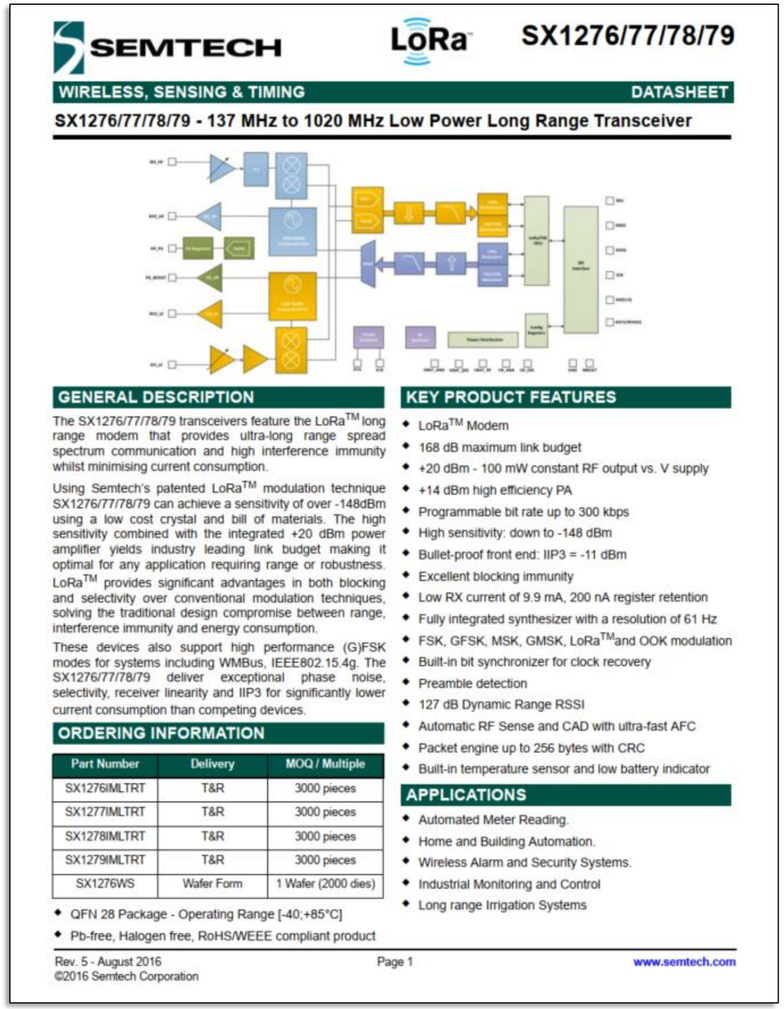
$$R_s = \frac{BW}{2^{SF}}\ simb/seg \qquad T_s = \frac{2^{SF}}{BW}\ seg$$

Tempo do preâmbulo (seg) – eq. 1:

$$T_{preamble} = \left(n_{preamble} + 4{,}25\right).T_s$$

ToA - Tempo de envio do pacote (seg) - eq. 2:

$$T_{packet} = T_{preamble} + T_{payload}$$



SEMTECH LoRa SX1276/77/78/79

WIRELESS, SENSING & TIMING DATASHEET

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Parâmetros temporais

Símbolos por payload - eq. 3:

$$n_{payload} = 8 + \max\left\{ceil\left[\frac{8PL - 4SF + 28 + 16CRC - 20IH}{4(SF - 2DE)}\right](CR + 4), 0\right\}$$

Tempo para envio do pacote - eq. 4:

$$T_{payload} = n_{payload} \times T_s$$

Parâmetros temporais

- Calculadora LoRa – SEMTECH:

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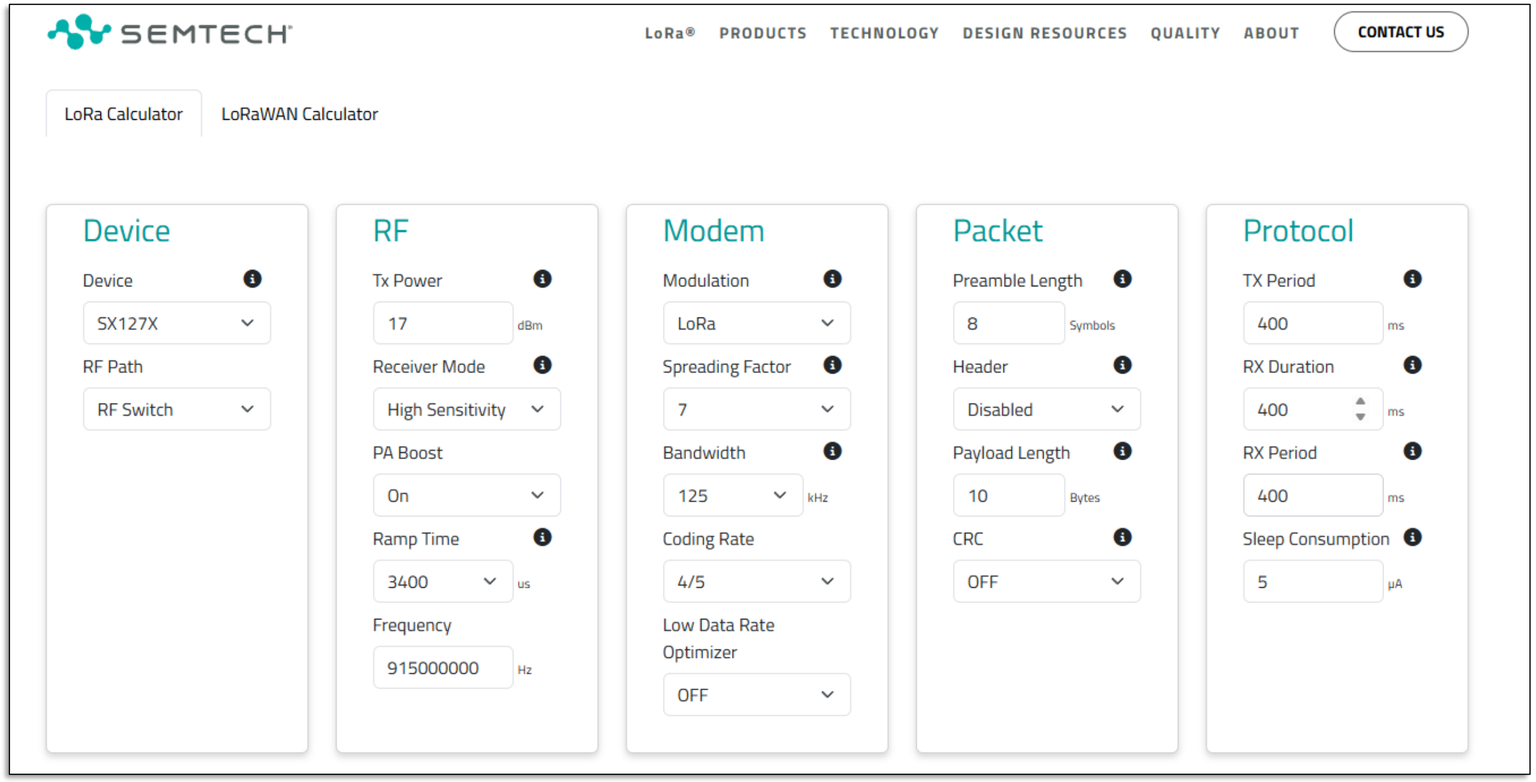

https://www.semtech.com/design-support/lora-calculator

Parâmetros temporais

- Calculadora LoRa – SEMTECH:

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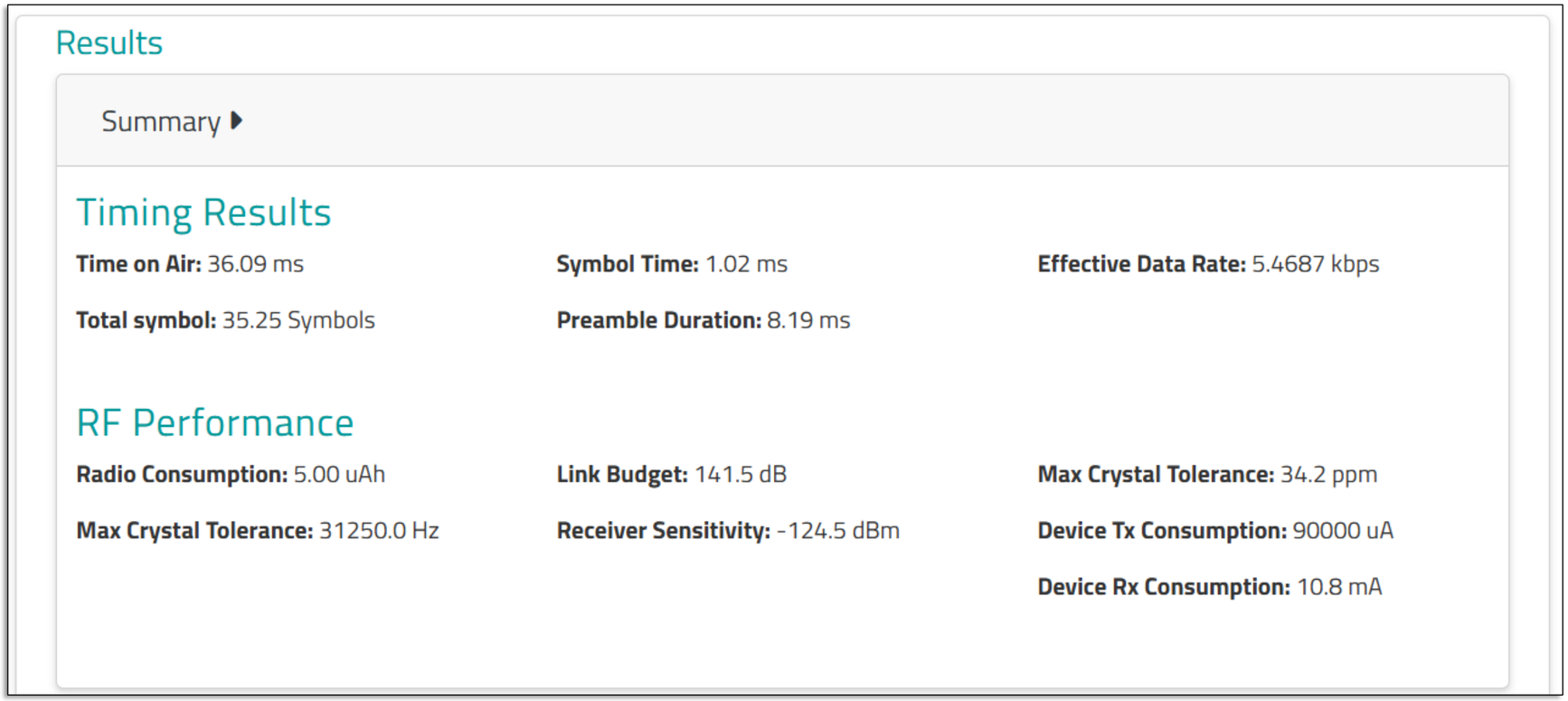


Results

Summary

Timing Results

Time on Air: 36.09 ms

Symbol Time: 1.02 ms

Effective Data Rate: 5.4687 kbps

Total symbol: 35.25 Symbols

Preamble Duration: 8.19 ms

RF Performance

Radio Consumption: 5.00 uAh

Link Budget: 141.5 dB

Max Crystal Tolerance: 34.2 ppm

Max Crystal Tolerance: 31250.0 Hz

Receiver Sensitivity: -124.5 dBm

Device Tx Consumption: 90000 uA

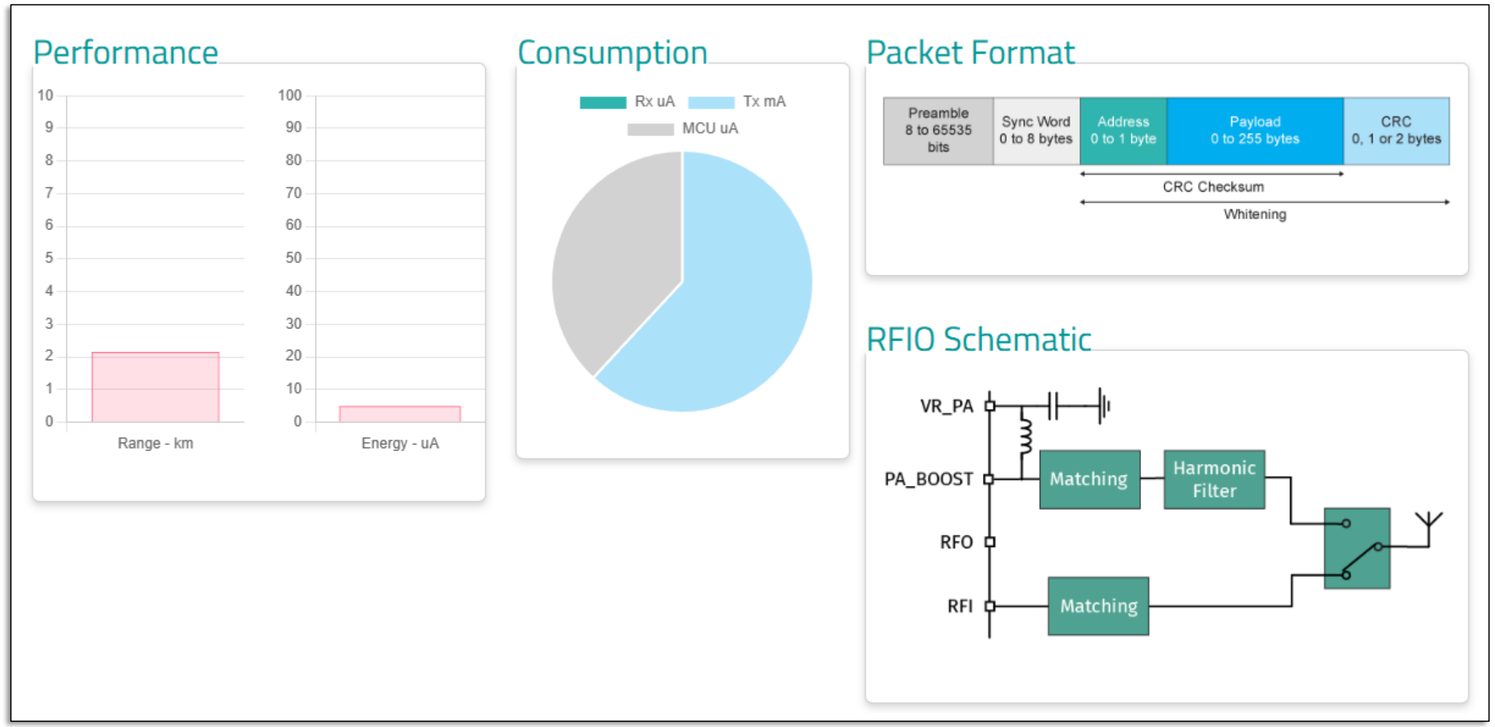
Device Rx Consumption: 10.8 mA

https://www.semtech.com/design-support/lora-calculator

Parâmetros temporais

- Calculadora LoRa – SEMTECH:

.



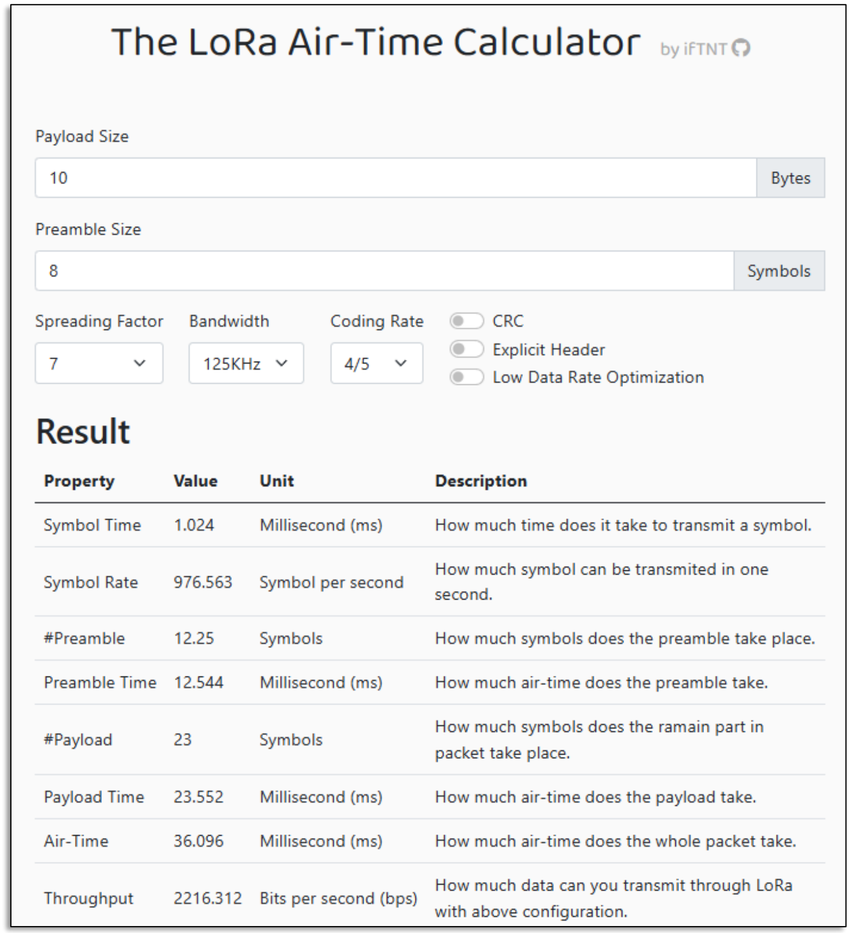
https://www.semtech.com/design-support/lora-calculator

Parâmetros temporais

- Exemplo – veja os dados presentes na figura ao lado:

.

$$T_s = \frac{2^{SF}}{BW} = \frac{2^7}{125.000} = 1{,}024\ ms$$



Property	Value	Unit	Description
Symbol Time	1.024	Millisecond (ms)	How much time does it take to transmit a symbol.
Symbol Rate	976.563	Symbol per second	How much symbol can be transmited in one second.
#Preamble	12.25	Symbols	How much symbols does the preamble take place.
Preamble Time	12.544	Millisecond (ms)	How much air-time does the preamble take.
#Payload	23	Symbols	How much symbols does the ramain part in packet take place.
Payload Time	23.552	Millisecond (ms)	How much air-time does the payload take.
Air-Time	36.096	Millisecond (ms)	How much air-time does the whole packet take.
Throughput	2216.312	Bits per second (bps)	How much data can you transmit through LoRa with above configuration.

https://iftnt.github.io/lora-air-time/index.html

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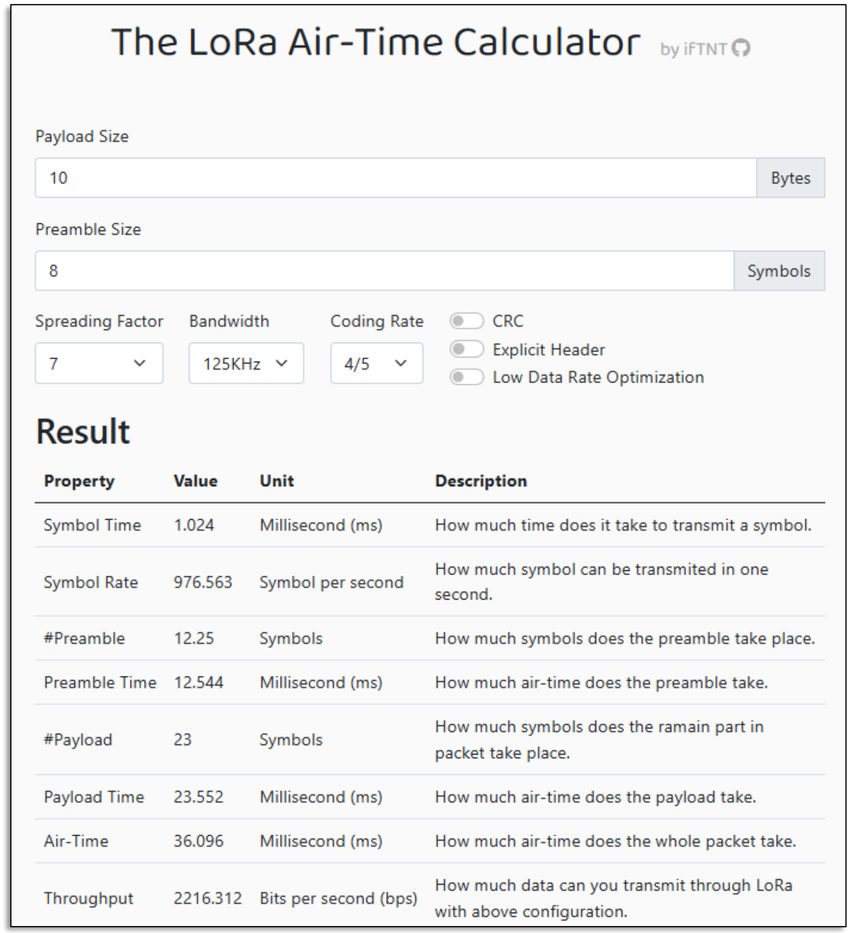

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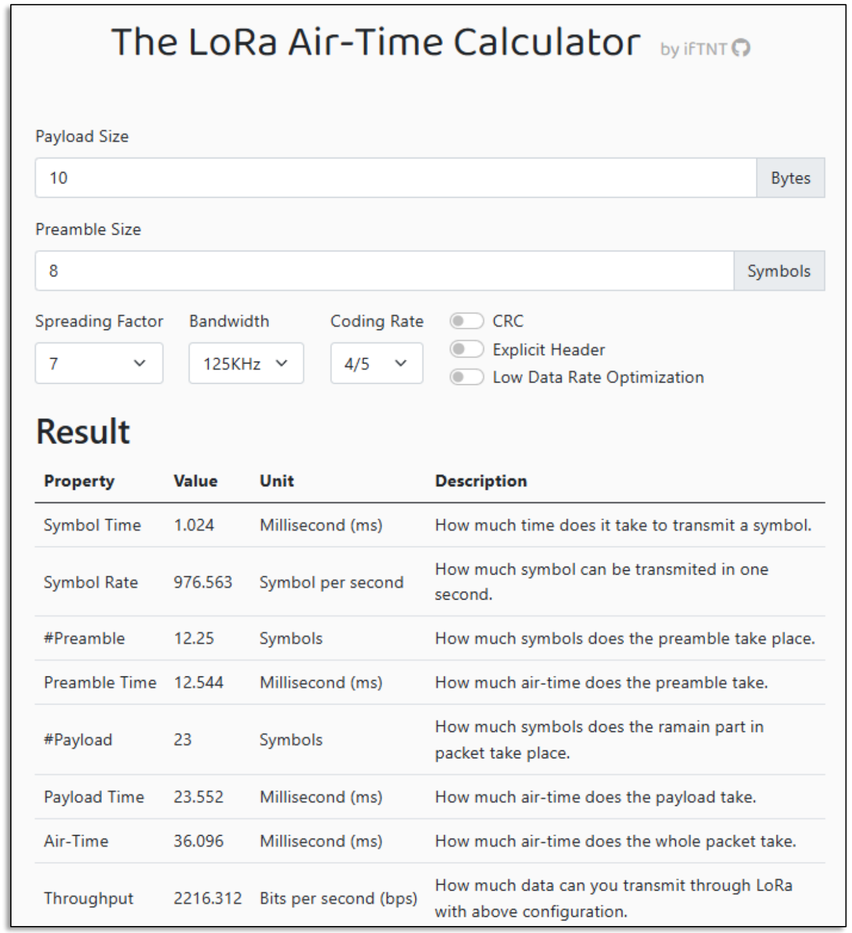
$$n_{payload} = 8 + \max\{ceil[2{,}857] \times (5), 0\}$$

3



The LoRa Air-Time Calculator by iFTNT

Payload Size

10 Bytes

Preamble Size

8 Symbols

Spreading Factor: 7

Bandwidth: 125KHz

Coding Rate: 4/5

CRC

Explicit Header

Low Data Rate Optimization

Result

Property	Value	Unit	Description
Symbol Time	1.024	Millisecond (ms)	How much time does it take to transmit a symbol.
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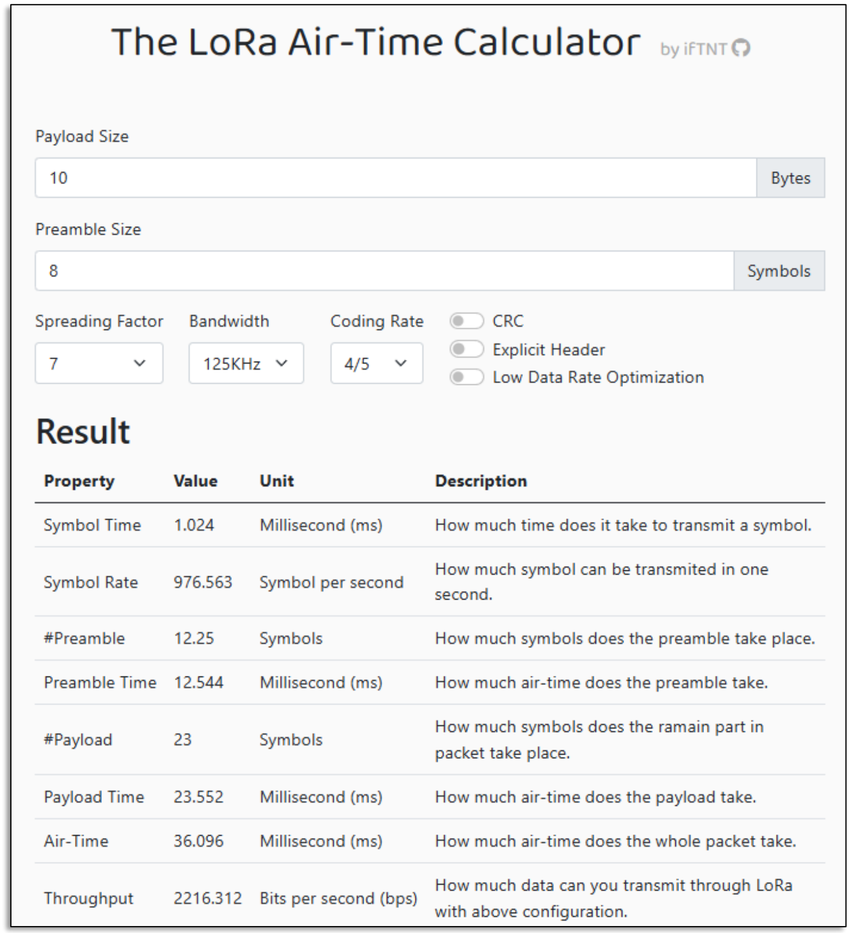

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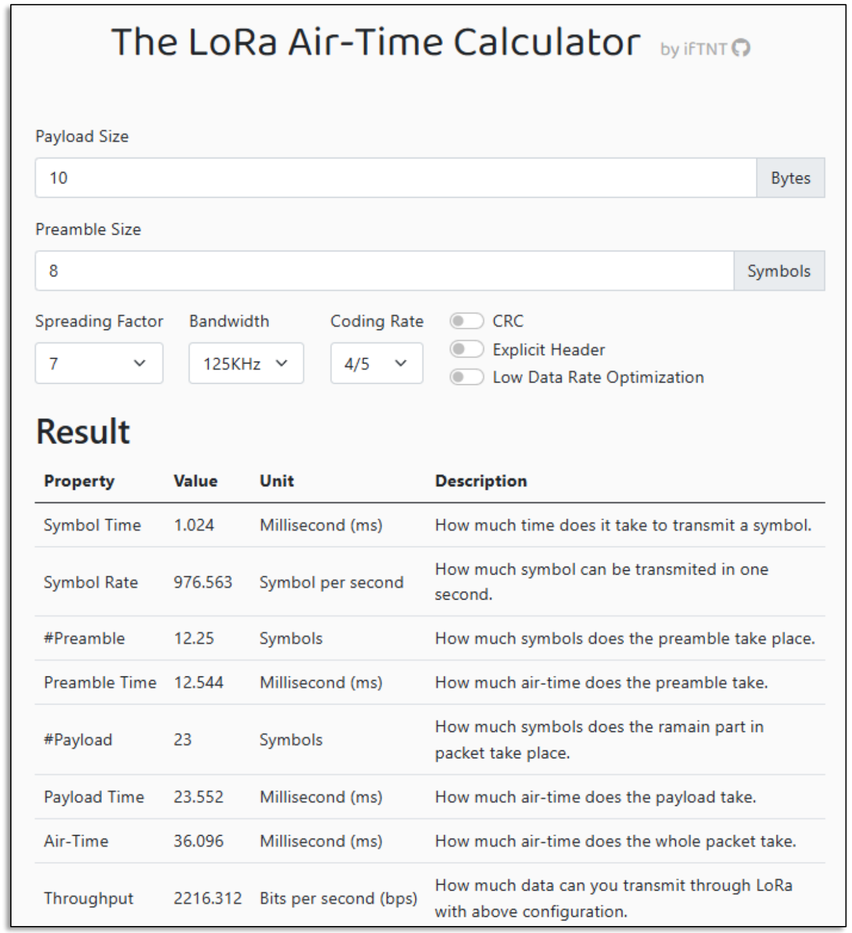

The LoRa Air-Time Calculator by iFTNT

Payload Size: 10 Bytes

Preamble Size: 8 Symbols

Spreading Factor: 7 | Bandwidth: 125KHz | Coding Rate: 4/5

CRC
Explicit Header
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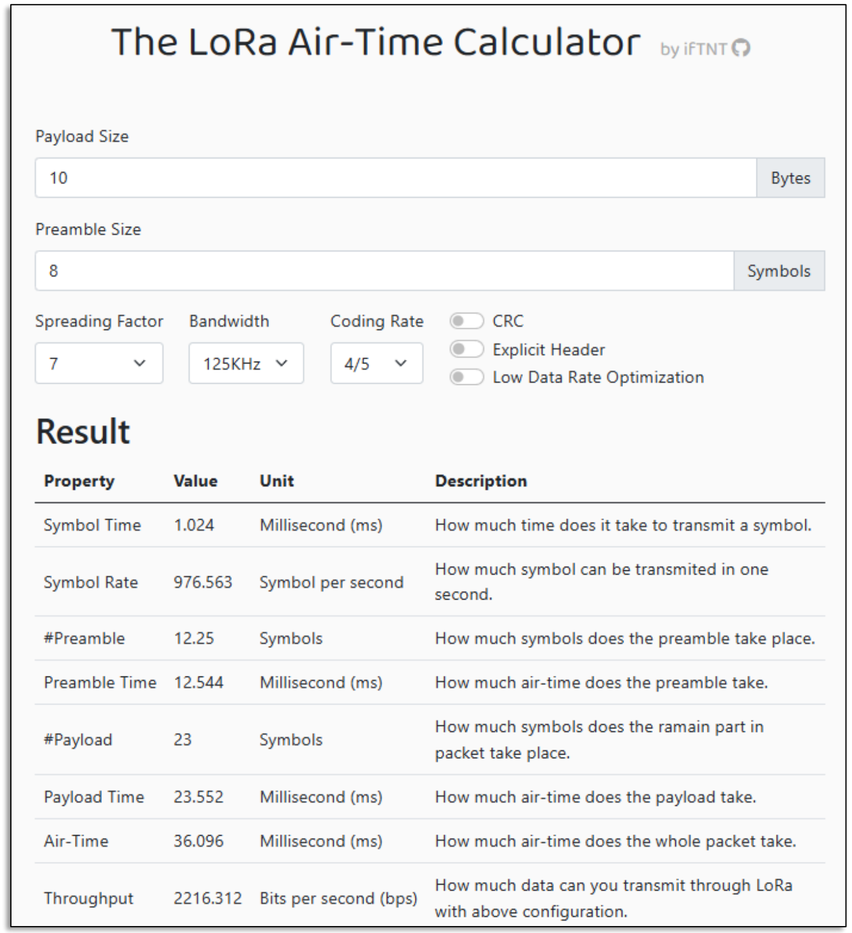
$$T_{packet} = T_{preamble} + T_{payload} = 36{,}096\ ms$$



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https://iftnt.github.io/lora-air-time/index.html

Parâmetros temporais

- Exemplo C – trecho do código:

.

```
// Cálculo de Rs e Ts
    symbol_rate = (bw)/pow(2, sf);
    symbol_period = 1/symbol_rate;
    printf("Rs: %.02f sim/s, Ts: %f ms \n", symbol_rate, symbol_period*1000);

    // Bit Rate (Rb)
    bit_rate = (sf * (4.0 / (4.0 + cr))) / (pow(2, sf) / bw);
    printf("Rb: %f bps\n", bit_rate);

    // Tempo do preâmbulo e do payload
    t_preamble = (preamble_len + 4.25)*symbol_period;
    n_payload = 8 + MAX(ceil((8.0*payload_len - 4.0*sf + 28.0 - 16.0*crc - 20.0*ih)/(4.0*(sf-2.0*de)))*(cr+4.0) , 0);
    t_payload = (float)n_payload*symbol_period;
    printf("Preamble length: %f ms e o payload length %f ms \n", t_preamble*1000, t_payload*1000);

    // Tempo do Time on Air (ToM)
    toa = t_preamble+t_payload;
    printf("ToA: %f ms\n", toa*1000);
```

Monitor Serial:

```
---- Porta serial reaberta COM7 ----
Rs: 976.56 sim/s, Ts: 1.024000 ms
Rb: 5468.750000 bps
Preamble length: 12.544000 ms e
o payload length 23.552000 ms
ToA: 36.096001 ms
---- A porta serial COM7 fechou ----
```

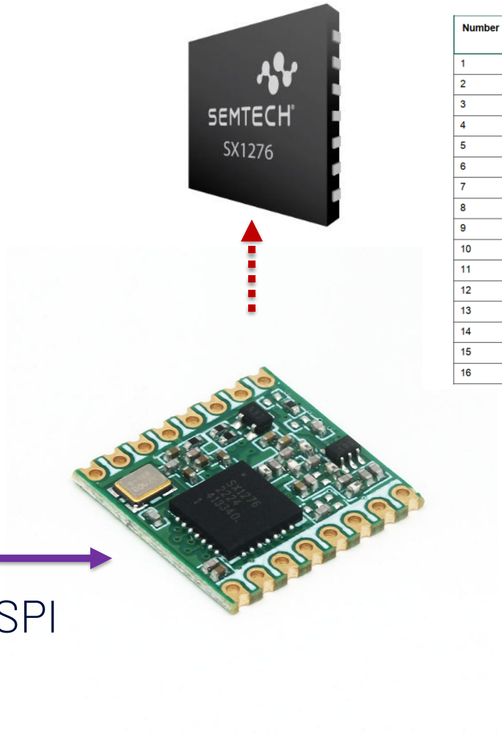
SX1276 e RFM95W

- Periférico da BitDogLab:

.

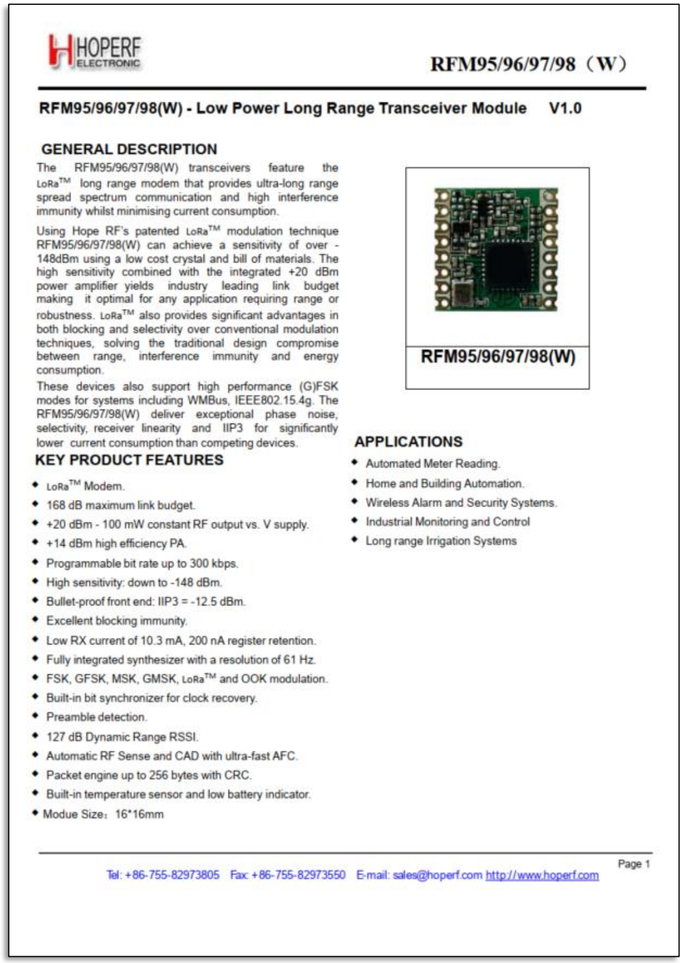
BitDogLAB

SPI



1.4. Pin Description

Number	Name	Type	Description Description Stand Alone Mode
1	GND	-	Ground
2	MISO	I	SPI Data output
3	MOSI	O	SPI Data input
4	SCK	I	SPI Clock input
5	NSS	I	SPI Chip select input
6	RESET	I/O	Reset trigger input
7	DIO5	I/O	Digital I/O, software configured
8	GND	-	Ground
9	ANT	-	RF signal output/input.
10	GND	-	Ground
11	DIO3	I/O	Digital I/O, software configured
12	DIO4	I/O	Digital I/O, software configured
13	3.3V	-	Supply voltage
14	DIO0	I/O	Digital I/O, software configured
15	DIO1	I/O	Digital I/O, software configured
16	DIO2	I/O	Digital I/O, software configured



HOPERF ELECTRONIC

RFM95/96/97/98（W）

RFM95/96/97/98(W) - Low Power Long Range Transceiver Module V1.0

GENERAL DESCRIPTION

The RFM95/96/97/98(W) transceivers feature the LoRa™ long range modem that provides ultra-long range spread spectrum communication and high interference immunity whilst minimising current consumption.

Using Hope RF's patented LoRa™ modulation technique RFM95/96/97/98(W) can achieve a sensitivity of over -148dBm using a low cost crystal and bill of materials. The high sensitivity combined with the integrated +20 dBm power amplifier yields industry leading link budget making it optimal for any application requiring range or robustness. LoRa™ also provides significant advantages in both blocking and selectivity over conventional modulation techniques, solving the traditional design compromise between range, interference immunity and energy consumption.

These devices also support high performance (G)FSK modes for systems including WMBus, IEEE802.15.4g. The RFM95/96/97/98(W) deliver exceptional phase noise, selectivity, receiver linearity and IIP3 for significantly lower current consumption than competing devices.

RFM95/96/97/98(W)

KEY PRODUCT FEATURES

- LoRa™ Modem.
- 168 dB maximum link budget.
- +20 dBm - 100 mW constant RF output vs. V supply.
- +14 dBm high efficiency PA.
- Programmable bit rate up to 300 kbps.
- High sensitivity: down to -148 dBm.
- Bullet-proof front end: IIP3 = -12.5 dBm.
- Excellent blocking immunity.
- Low RX current of 10.3 mA, 200 nA register retention.
- Fully integrated synthesizer with a resolution of 61 Hz.
- FSK, GFSK, MSK, GMSK, LoRa™ and OOK modulation.
- Built-in bit synchronizer for clock recovery.
- Preamble detection.
- 127 dB Dynamic Range RSSI.
- Automatic RF Sense and CAD with ultra-fast AFC.
- Packet engine up to 256 bytes with CRC.
- Built-in temperature sensor and low battery indicator.
- Modue Size：16*16mm

APPLICATIONS

- Automated Meter Reading.
- Home and Building Automation.
- Wireless Alarm and Security Systems.
- Industrial Monitoring and Control
- Long range Irrigation Systems

Tel: +86-755-82973805 Fax: +86-755-82973550 E-mail: sales@hoperf.com http://www.hoperf.com

Page 1

SX1276 e RFM95W

- Periférico da BitDogLab:

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BitDogLAB

SPI

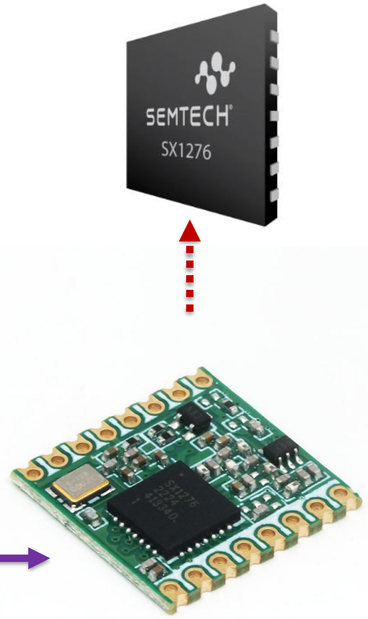
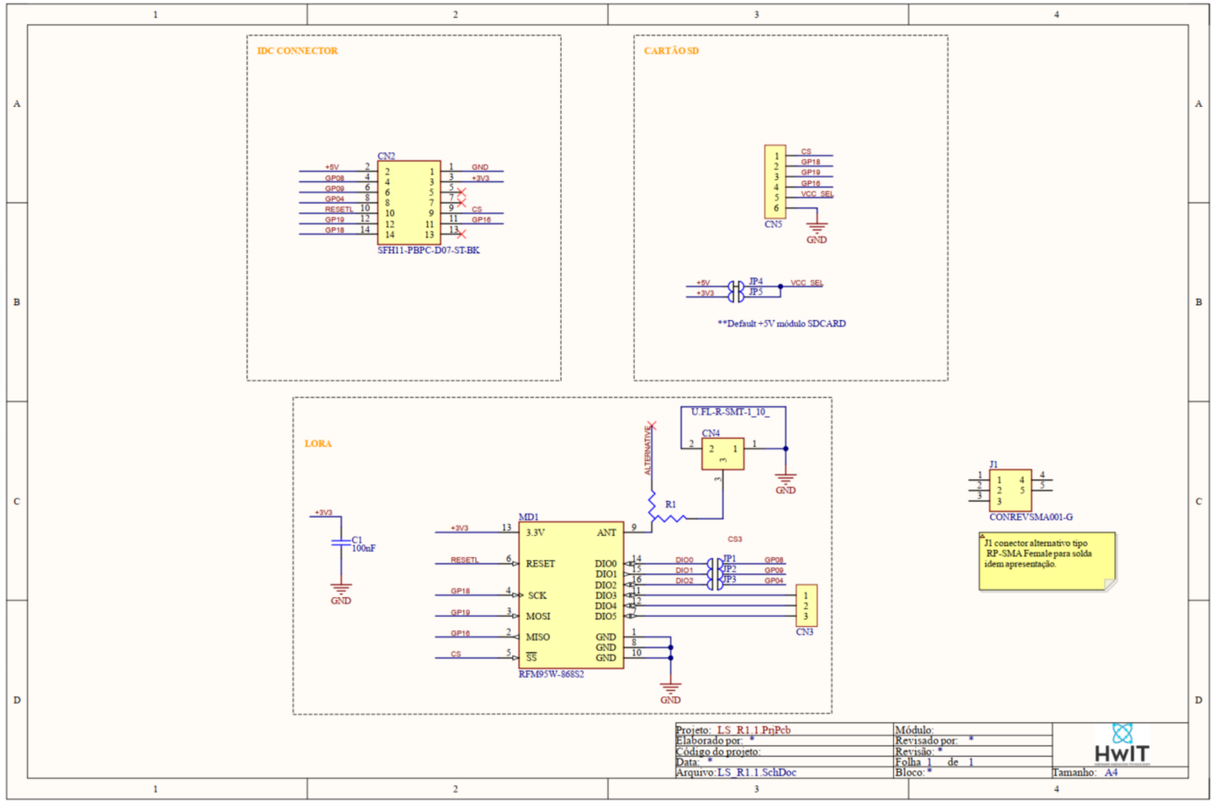


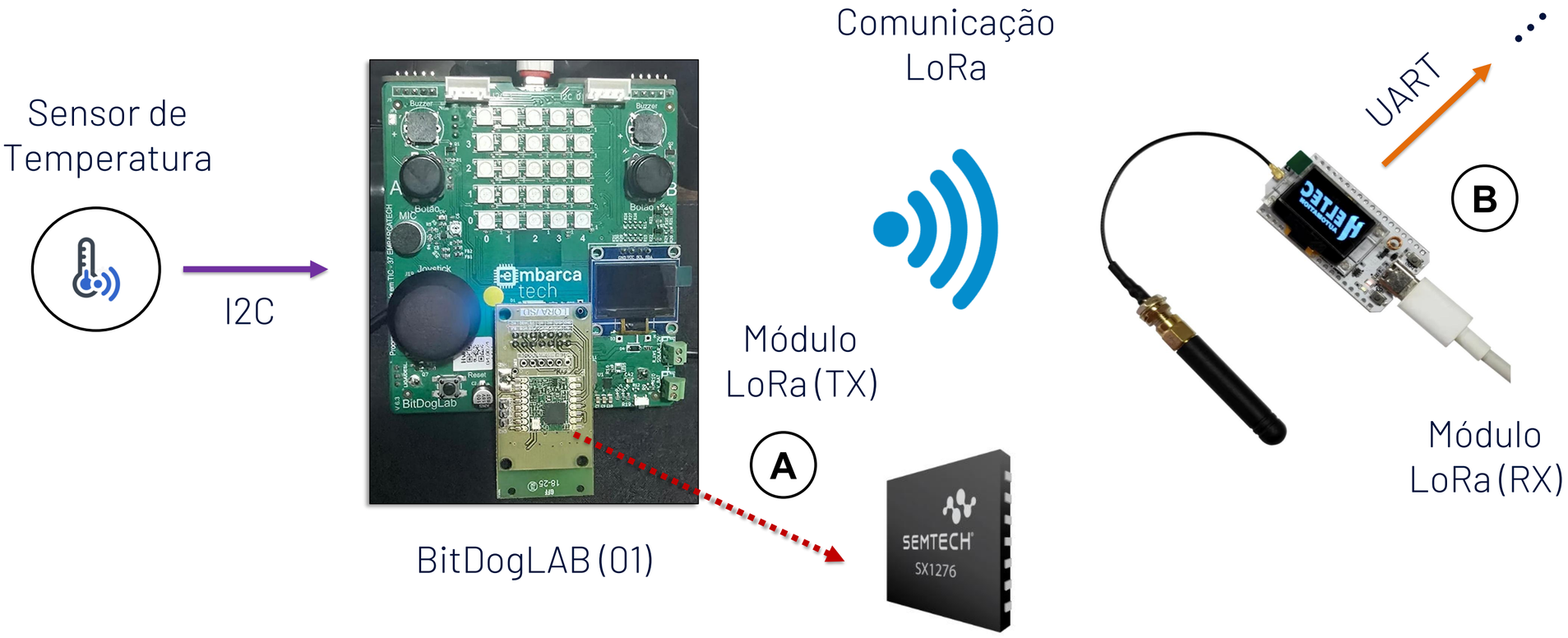
Diagrama esquemático do periférico BitDogLab



Prática - Comunicação LoRa

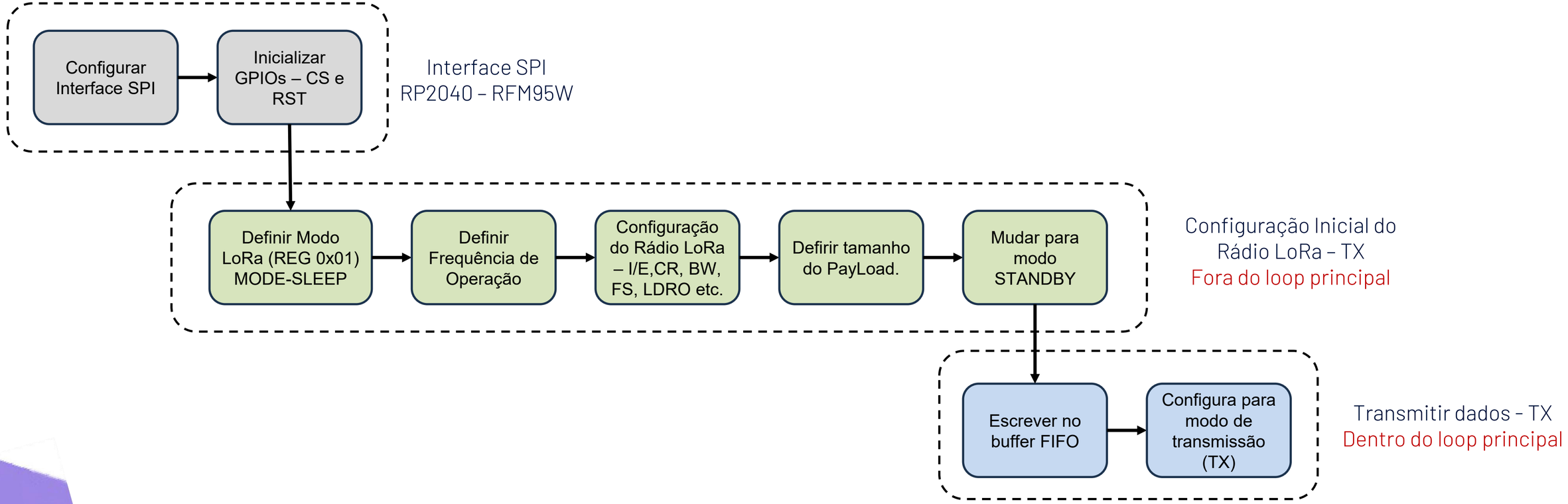
Prática – comunicação LoRa

- Experimento LoRa:



Prática – comunicação LoRa

- Etapas para a construção do código do transmissor LoRa (TX):



Prática – comunicação LoRa

- Transmissor LoRa (TX) – orientações e dicas:

```
// Configurações do - TX:
int payload_len = 10;       // Tamanho do payload em byte
int preamble_len = 8;       // Tamanho do preâmbulo
int sf = 7;                 // fator de espalhamento (Spreading Factor)
int crc = 0;                // tamanho do CRC (Cyclic Redundancy Check) em bytes (normalmente 2 bytes)
int ih = 0;                 // Cabeçalho implícito (Implicit Header) (0 ou 1)
int bw = 125000;            // largura de banda (Bandwidth) em Hz
int cr = 1;                 // Taxa de codificação (Coding Rate)
int de = 0;                 //  LowDataRateOptimize=1, DE=0 caso contrário
```

Frequência escolhida 915.000 MHz

Frequencia calculada para ajuste dos registradores: 14991360
Impressão de valor binário da frequência:
00000000111001001100000000000000
Frequ. Regs. config:
READ - REG_FRF_MSB: E4
READ - REG_FRF_MID: C0
READ - REG_FRF_LSB: 00

Registradores SX1276

Name (Address)	Bits	Variable Name	Mode	Reset	LoRa™ Description
RegModemConfig1 (0x1D)	7-4	Bw	rw	0x07	Signal bandwidth: 0000 → 7.8 kHz 0001 → 10.4 kHz 0010 → 15.6 kHz 0011 → 20.8kHz 0100 → 31.25 kHz 0101 → 41.7 kHz 0110 → 62.5 kHz 0111 → 125 kHz 1000 → 250 kHz 1001 → 500 kHz other values → reserved In the lower band (169MHz), signal bandwidths 8&9 are not supported)
	3-1	CodingRate	rw	'001'	Error coding rate 001 → 4/5 010 → 4/6 011 → 4/7 100 → 4/8 All other values → reserved In implicit header mode should be set on receiver to determine expected coding rate. See 4.1.1.3
	0	ImplicitHeaderModeOn	rw	0x0	0 → Explicit Header mode 1 → Implicit Header mode
RegModemConfig2 (0x1E)	7-4	SpreadingFactor	rw	0x07	SF rate (expressed as a base-2 logarithm) 6 → 64 chips / symbol 7 → 128 chips / symbol 8 → 256 chips / symbol 9 → 512 chips / symbol 10 → 1024 chips / symbol 11 → 2048 chips / symbol 12 → 4096 chips / symbol other values reserved.
	3	TxContinuousMode	rw	0	0 → normal mode, a single packet is sent 1 → continuous mode, send multiple packets across the FIFO (used for spectral analysis)
	2	RxPayloadCrcOn	rw	0x00	Enable CRC generation and check on payload: 0 → CRC disable 1 → CRC enable If CRC is needed, RxPayloadCrcOn should be set: - in Implicit header mode: on Tx and Rx side - in Explicit header mode: on the Tx side alone (recovered from the header in Rx side)
	1-0	SymbTimeout(9:8)	rw	0x00	RX Time-Out MSB

Prática – comunicação LoRa

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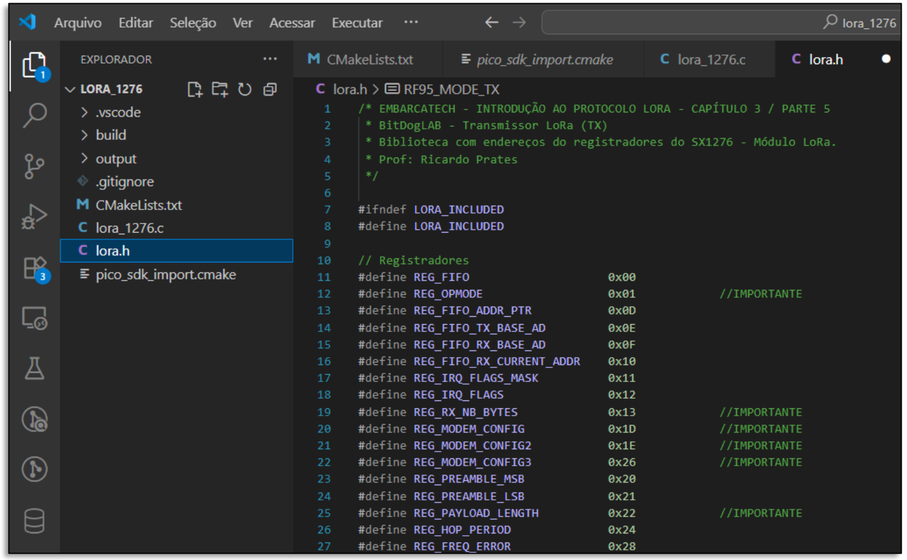
Biblioteca de registradores SX1276 – lora.h



Prática – comunicação LoRa

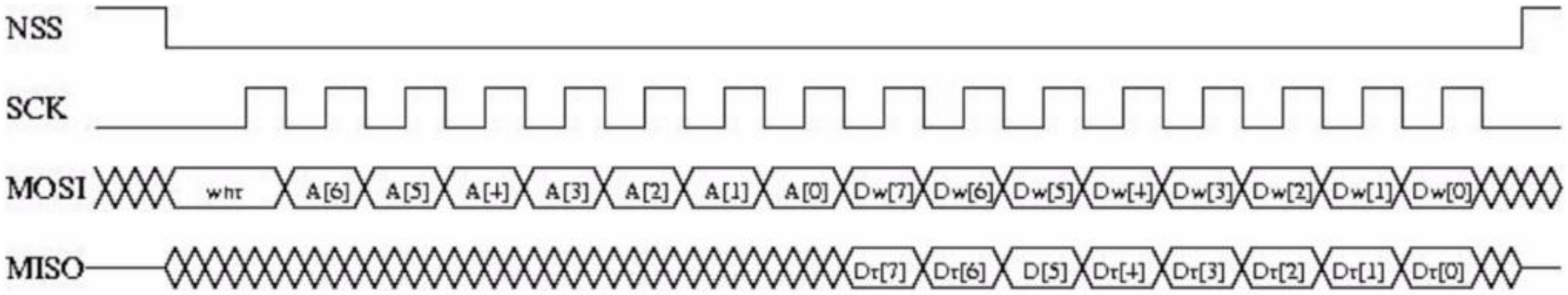
Exemplo de rotina para configurar a frequência de operação:

```
// Definir frequência de operação
void SetFrequency(double Frequency)
{
    unsigned long FrequencyValue;               //Frequência corrida - valor para os registradores do SX1276

    printf("Frequência escolhida %.3f MHz\n", Frequency);
    Frequency = Frequency * 7110656 / 434;      //Fórmula para correção da frequência - considerando o padrão de 434 MHz - 6C 80 00
    FrequencyValue = (unsigned long)(Frequency);
    printf("Frequencia calculada para ajuste dos registradores: %lu\n", FrequencyValue);

    writeRegister(REG_FRF_MSB, (FrequencyValue >> 16) & 0xFF);  // Escrita no registrador RegFrMsb  (0x06)
    writeRegister(REG_FRF_MID, (FrequencyValue >> 8) & 0xFF);   // Escrita no registrador RegFrMid  (0x07)
    writeRegister(REG_FRF_LSB, FrequencyValue & 0xFF);          // Escrita no registrador RegFrLsb  (0x08)

    printf("Impressão de valor binário da frequência:\n");
    imprimir_binario(FrequencyValue);                           // Imprimir o valor binário de 32 bits - FrequencyValue
    printf("\n");

    printf("Frequ. Regs. config:\n");
    // Teste de leitura dos registradores RegFrMsb, RegFrMid e RegFrLsb.
    readRegister(REG_FRF_MSB);
    readRegister(REG_FRF_MID);
    readRegister(REG_FRF_LSB);
}
```

```
00000000 11100100 11000000 00000000
          0xE4     0xC0     0x00
```

Prática – comunicação LoRa

Emprego do protocolo SPI no SX1276 – 4.3.:



The first byte is the address byte. It is comprises:

- A wnr bit, which is 1 for write access and 0 for read access.
- Then 7 bits of address, MSB first.

Prática – comunicação LoRa

Exemplo de rotina para escrever no registrador LoRa:

```
// Escrever no registrador do Rádio LoRa - SX1276
static void writeRegister(uint8_t reg, uint8_t data)
{
    uint8_t buf[2];                         // Buffer para escrita no módulo LoRa

    buf[0] = reg | 0x80;                    // Registrador com operação de escrita
    buf[1] = data;                          // Byte de informação inserida no registrador

    cs_select();                            // Iniciar comunicação SPI
    spi_write_blocking(SPI_PORT, buf, 2);   // Transmitir informação do Buffer pela interface SPI
    cs_deselect();                          // Finalizar comunicação SPI

    sleep_ms(1);                            // Delay de 1 ms
}
```

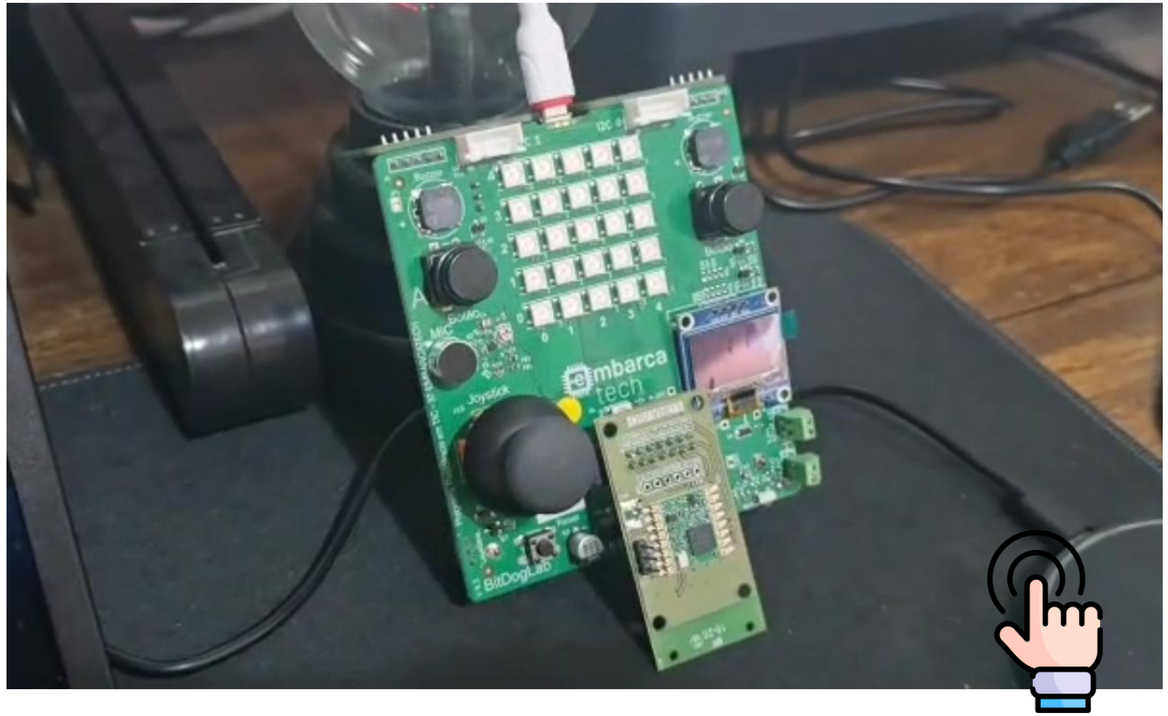
Prática – comunicação LoRa

Exemplo de rotina para ler um determinado registrador LoRa:

```
// Ler conteúdo registrador do Rádio LoRa - SX1276
static uint8_t readRegister(uint8_t addr)
{
    uint8_t buf[1];                          // Buffer para leitura do módulo LoRa

    addr &= 0x7F;                            // Resgistrador com operação de leitura

    cs_select();                             // Iniciar comunicação SPI
    spi_write_blocking(SPI_PORT, &addr, 1); // Transmitir informação do Buffer pela interface SPI - Endereço de leitura
    sleep_ms(1);                             // Delay de 1 ms
    spi_read_blocking(SPI_PORT, 0, buf, 1); // Receber informação pela interface SPI e armazenar no buffer
    cs_deselect();                           // Finalizar comunicação SPI

    sleep_ms(1);                             // Delay de 1 ms

    printf("READ %02X\n", buf[0]);           // Imprimir resultado obtido pela leitura do registrador
    return buf[0];                           // Retornar resultado do valor armazenado no registrador
}
```

Prática – comunicação LoRa

Vídeo com exemplo de funcionamento:



Monitor Serial:

```
---- Porta serial reaberta COM7 ----
Iniciando o rádio LoRa
Frequência escolhida 915.000 MHz
Frequencia calculada para ajuste dos registradores: 14991360
Impressão de valor binário da frequência:
00000000111001001100000000000000
Frequ. Regs. config:
READ E4
READ C0
READ 00
RegDetectOp:
READ C3
Configuração LoRa completa
Rs: 976.56 sim/s, Ts: 1.024000 ms
Rb: 5468.750000 bps
Preamble length: 12.544000 ms e o payload length 23.552000 ms
ToA: 36.096001 ms
Mensagem enviada: 28.40
Mensagem enviada: 22.90
Mensagem enviada: 26.40
---- A porta serial COM7 fechou ----
```

Conclusões

Conclusões

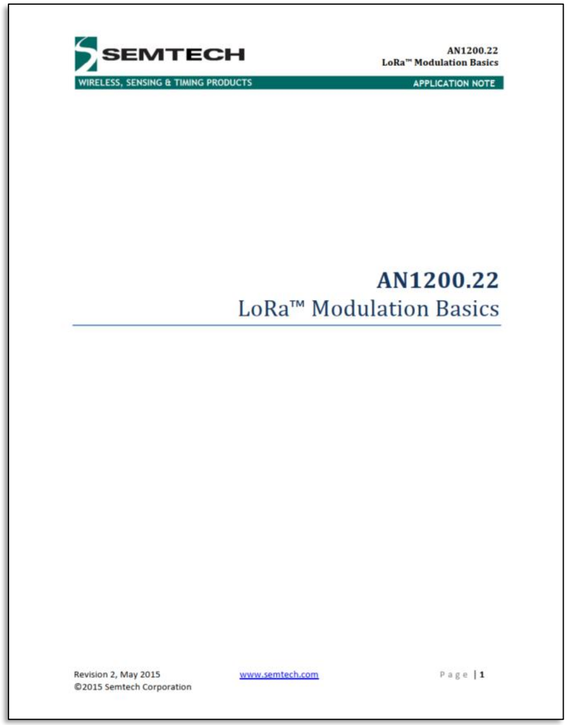
- A interface LoRa possui uma série parâmetros que podem ser ajustados para otimizar a transmissão de pacotes de dados.

- A estrutura de pacotes da comunicação LoRa é bastante flexível e eficiente para diferentes cenários, principalmente na presença de obstáculos e grandes distâncias.

- O transceptor SX1276 e o módulo RFM95W podem se comunicar, de forma bastante eficiente, com o microcontrolador RP2040 através da interface SPI.

- É possível configurar os registradores do SX1276 e desenvolver um rádio transmissor LoRa (Tx) através da ferramenta educacional BitDogLab, juntamente com o emprego da linguagem C e o Kit de Desenvolvimento de Software Pico SDK.

Bibliografia Recomendada



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AN1200.22
LoRa™ Modulation Basics

WIRELESS, SENSING & TIMING PRODUCTS — APPLICATION NOTE

AN1200.22
LoRa™ Modulation Basics

Revision 2, May 2015
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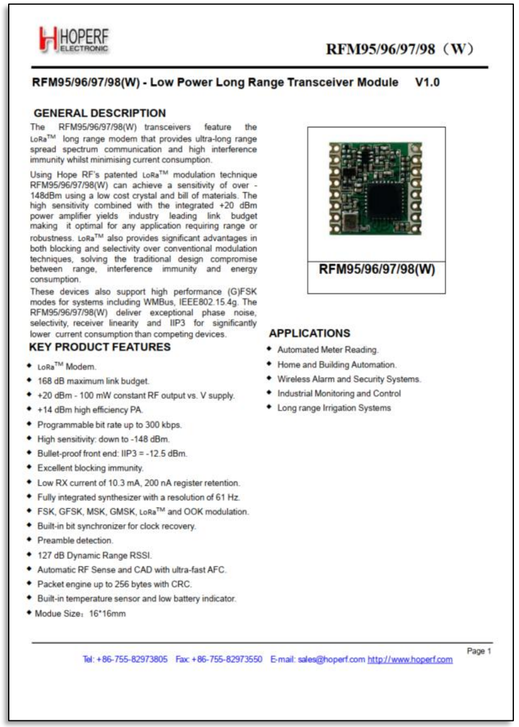

HOPERF ELECTRONIC — RFM95/96/97/98（W）

RFM95/96/97/98(W) - Low Power Long Range Transceiver Module V1.0

GENERAL DESCRIPTION

The RFM95/96/97/98(W) transceivers feature the LoRa™ long range modem that provides ultra-long range spread spectrum communication and high interference immunity whilst minimising current consumption.

Using Hope RF's patented LoRa™ modulation technique RFM95/96/97/98(W) can achieve a sensitivity of over -148dBm using a low cost crystal and bill of materials. The high sensitivity combined with the integrated +20 dBm power amplifier yields industry leading link budget making it optimal for any application requiring range or robustness. LoRa™ also provides significant advantages in both blocking and selectivity over conventional modulation techniques, solving the traditional design compromise between range, interference immunity and energy consumption.

These devices also support high performance (G)FSK modes for systems including WMBus, IEEE802.15.4g. The RFM95/96/97/98(W) deliver exceptional phase noise, selectivity, receiver linearity and IIP3 for significantly lower current consumption than competing devices.

KEY PRODUCT FEATURES

- LoRa™ Modem.
- 168 dB maximum link budget.
- +20 dBm - 100 mW constant RF output vs. V supply.
- +14 dBm high efficiency PA.
- Programmable bit rate up to 300 kbps.
- High sensitivity: down to -148 dBm.
- Bullet-proof front end: IIP3 = -12.5 dBm.
- Excellent blocking immunity.
- Low RX current of 10.3 mA, 200 nA register retention.
- Fully integrated synthesizer with a resolution of 61 Hz.
- FSK, GFSK, MSK, GMSK, LoRa™ and OOK modulation.
- Built-in bit synchronizer for clock recovery.
- Preamble detection.
- 127 dB Dynamic Range RSSI.
- Automatic RF Sense and CAD with ultra-fast AFC.
- Packet engine up to 256 bytes with CRC.
- Built-in temperature sensor and low battery indicator.
- Modue Size：16*16mm

RFM95/96/97/98(W)

APPLICATIONS

- Automated Meter Reading.
- Home and Building Automation.
- Wireless Alarm and Security Systems.
- Industrial Monitoring and Control
- Long range Irrigation Systems

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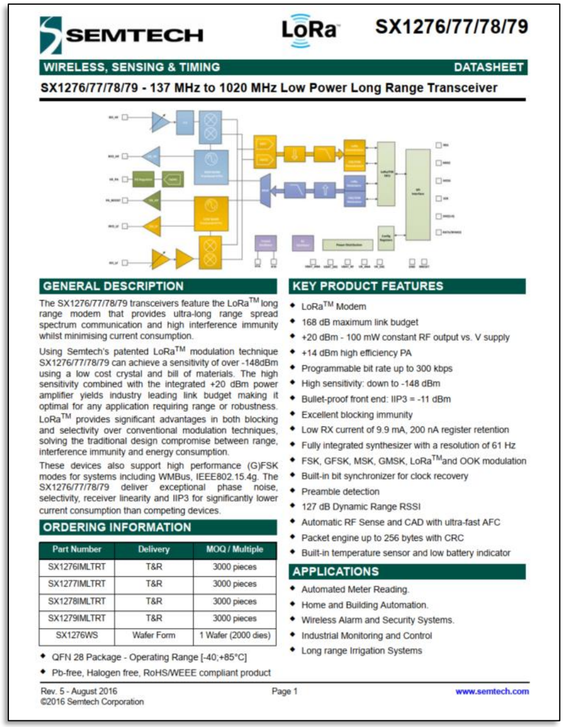

SEMTECH — LoRa — SX1276/77/78/79

WIRELESS, SENSING & TIMING — DATASHEET

SX1276/77/78/79 - 137 MHz to 1020 MHz Low Power Long Range Transceiver

GENERAL DESCRIPTION

The SX1276/77/78/79 transceivers feature the LoRa™ long range modem that provides ultra-long range spread spectrum communication and high interference immunity whilst minimising current consumption.

Using Semtech's patented LoRa™ modulation technique SX1276/77/78/79 can achieve a sensitivity of over -148dBm using a low cost crystal and bill of materials. The high sensitivity combined with the integrated +20 dBm power amplifier yields industry leading link budget making it optimal for any application requiring range or robustness. LoRa™ provides significant advantages in both blocking and selectivity over conventional modulation techniques, solving the traditional design compromise between range, interference immunity and energy consumption.

These devices also support high performance (G)FSK modes for systems including WMBus, IEEE802.15.4g. The SX1276/77/78/79 deliver exceptional phase noise, selectivity, receiver linearity and IIP3 for significantly lower current consumption than competing devices.

ORDERING INFORMATION

Part Number	Delivery	MOQ / Multiple
SX1276IMLTRT	T&R	3000 pieces
SX1277IMLTRT	T&R	3000 pieces
SX1278IMLTRT	T&R	3000 pieces
SX1279IMLTRT	T&R	3000 pieces
SX1276WS	Wafer Form	1 Wafer (2000 dies)

- QFN 28 Package - Operating Range [-40;+85°C]
- Pb-free, Halogen free, RoHS/WEEE compliant product

KEY PRODUCT FEATURES

- LoRa™ Modem
- 168 dB maximum link budget
- +20 dBm - 100 mW constant RF output vs. V supply
- +14 dBm high efficiency PA
- Programmable bit rate up to 300 kbps
- High sensitivity: down to -148 dBm
- Bullet-proof front end: IIP3 = -11 dBm
- Excellent blocking immunity
- Low RX current of 9.9 mA, 200 nA register retention
- Fully integrated synthesizer with a resolution of 61 Hz
- FSK, GFSK, MSK, GMSK, LoRa™and OOK modulation
- Built-in bit synchronizer for clock recovery
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Comunicação em IoT – Parte 5

RESIDÊNCIA – Aula Síncrona (05/08/2025)

Prof. Dr Ricardo Menezes Prates

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